HARROWMERE WATER AUTHORITY

Asset Management Plan 2027–2036

Ten-year plan for treatment, network and storage assets

Version 3.1 · Submitted to the Board for approval
Prepared by the Asset Strategy Group

Synthetic sample. Harrowmere Water Authority is a fictional organisation. All places, names and figures in this document are invented for software testing.

1 Introduction

1.1 Purpose of this plan

This plan sets out how the Authority will look after the treatment works, pipes, pumping stations and service reservoirs that supply drinking water to around 640,000 people over the ten years from April 2027. It explains what condition the assets are in, which risks they pose to customers, and which interventions are proposed to manage those risks at the lowest whole-life cost.

The plan is the main input to the Authority’s business plan for the next two regulatory periods. It is also the reference for the annual investment programme, which turns the interventions described here into individual projects with their own budgets, approvals and delivery dates.

The Board is asked to approve the plan as the basis for that business plan. Approval does not commit expenditure on any individual project; each project still needs approval at the gate set out in the Authority’s investment governance framework.

1.2 Scope

The plan covers all assets used to abstract, treat, store and distribute drinking water, from the raw water intakes to the boundary of each customer’s property. It includes the communication pipes that the Authority owns, but not the supply pipes inside private land, which remain the responsibility of property owners.

Non-infrastructure assets that support operations, such as depots, vehicles, laboratories and office buildings, are covered by the separate corporate property plan. Information technology systems are covered only where they monitor or control operational assets, in particular telemetry, the control room and the hydraulic models.

The plan does not cover wastewater, which in this region is the responsibility of a separate body. Where water and wastewater works share a site or a route, the plan notes the dependency in the relevant zone profile.

1.3 How the plan was prepared

The plan was prepared by the Asset Strategy Group over eighteen months. It draws on the asset register, condition surveys of all 61 service reservoirs, burst and interruption records for the last ten years, the leakage model, and workshops with operations staff in each of the four service areas.

Each zone was assessed on the same basis: the likelihood that an asset fails within the planning period, the consequence of that failure for customers and the environment, and the cost of the available options to reduce the risk. Options were compared on whole-life cost over forty years, using the discount rate set by the regulator.

A draft of the plan was published for consultation between May and July 2026. The Authority received 214 responses from customers, parish councils, developers and other utilities. The main themes were the pace of lead pipe replacement, the disruption caused by street works and the resilience of supplies to rural villages.

The figures in this version reflect the asset register as at 31 March 2026 and prices at 2026/27 levels. Where later information was available, for example from surveys completed in the autumn of 2026, it is mentioned in the text but has not been used to change the costed programme.

1.4 Structure of the document

Chapter 2 describes the strategic context: demand, regulation, climate and affordability. Chapter 3 summarises the asset base. Chapter 4 describes the nine treatment works, and Chapter 5 contains a profile of each of the twenty-four supply zones, with its assets, condition, risks and planned interventions. Chapters 6 and 7 bring the

interventions together into a capital programme and a financial plan. Chapter 8 describes the main risks to delivering the plan and how they are managed, and Chapter 9 explains how progress will be monitored. The appendices contain a glossary and the record of approval.

2 Strategic context

2.1 Demand outlook

Average demand across the region was 198 megalitres per day in 2025/26, of which about 62 per cent was household use. The population served is forecast to grow by 9 per cent over the planning period, mostly in the new settlements allocated in the local plans for the Fernlow and Quarrendon areas.

Household consumption per person has fallen slowly since meters became compulsory for new connections, and is now about 139 litres per person per day. The plan assumes that it falls further to 125 litres by 2036, supported by the metering programme, water efficiency visits and tariffs that reward lower use.

Peak demand during hot weather remains the main pressure on the network. On the hottest days of the 2025 summer, demand rose to 262 megalitres per day, and several reservoirs in the eastern zones fell below a quarter of their capacity overnight. The plan therefore gives priority to storage and transfer capacity in those zones.

Non-household demand is dominated by food processing and by two hospitals. The Authority has agreements with the largest users to reduce consumption during droughts, and the plan assumes that these agreements remain in place.

2.2 Regulatory obligations

The Authority must supply water that meets the drinking water quality regulations, maintain adequate pressure, and meet the performance commitments agreed with the economic regulator for leakage, interruptions, bursts and customer contacts. Failing a commitment leads to financial penalties and, in serious cases, enforcement action.

The drinking water inspectorate requires the Authority to complete the programme of lead communication pipe replacement agreed in its lead strategy, and to maintain service reservoirs so that they do not allow contamination. Two undertakings agreed with the inspectorate in 2024 are included in the programme: the refurbishment of the Gorsemere reservoirs and the replacement of the ultraviolet plant at Stavely Cross.

The environmental regulator sets abstraction licences and requires the Authority to reduce abstraction from the Elm Valley boreholes in dry years to protect river flows. The plan assumes that the licence change proposed for 2030 is confirmed.

2.3 Climate resilience

Climate projections for the region show hotter, drier summers and wetter winters with more intense rainfall. Both trends affect water assets: dry summers raise demand and shrink clay soils, which increases bursts; wet winters flood low-lying sites and raise the turbidity of river water.

The Authority has assessed the flood risk at all operational sites. Eleven sites are at risk from a flood with a one per cent annual probability, and four of them are critical. The plan includes flood protection at those four sites and the relocation of control kiosks above flood level at the others.

The plan also includes measures to keep supplies going during power cuts, which are expected to become more frequent during storms. All strategic pumping stations will have standby generation or a second, independent power supply by 2032.

2.4 Affordability

The investment in this plan would add about 4.1 per cent in real terms to the average household bill by 2036, before the effect of efficiency savings. The Authority's social tariff will continue to cap bills for households on low incomes at 3 per cent of household income.

To keep bills affordable, the plan phases the largest schemes so that expenditure is spread across both regulatory periods, and gives priority to interventions that reduce operating costs, such as pump replacement and pressure management.

3 Asset base overview

The table below summarises the main asset classes across all twenty-four supply zones, with their replacement value at 2026/27 prices. Network assets account for more than three quarters of the total value; treatment works and service reservoirs account for most of the rest.

Asset class	Quantity	Unit	Replacement value (£m)	Share of value (%)
Trunk mains	1,184	km	1,066	27.4
Distribution mains	6,912	km	1,935	49.8
Treatment works	9	no.	412	10.6
Service reservoirs	61	no.	238	6.1
Pumping stations	94	no.	131	3.4
Customer meters	214,600	no.	54	1.4
Telemetry and control	812	outstations	27	0.7
Other	–	–	23	0.6

The median age of the mains network is 54 years. About 18 per cent of distribution mains are cast iron laid before 1940, and these account for 46 per cent of bursts. The replacement rate in the last regulatory period was 0.4 per cent of the network per year; this plan raises it to 0.7 per cent, targeted at the mains with the highest burst rates.

Service reservoirs are on average 71 years old. The condition survey completed in 2025 graded 17 of them at condition grade 4 or 5. Most of the defects are in roofs and roof membranes, which are the main route for contamination, and in the internal coatings of the older concrete tanks.

4 Treatment works

The Authority operates nine treatment works and source sites. Together they can put up to 286 megalitres per day into supply, against a peak week demand of 248 megalitres per day in 2025. This chapter describes each site, the condition of its main process stages and the work planned for 2027–2036.

4.1 Brackwater works

Brackwater is a site for river abstraction and conventional treatment, built in 1958, with a maximum output of 74 megalitres per day. In 2025/26 it supplied an average of 60.5 megalitres per day and was out of service for 55 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Intake and screens	2	2001	Replacement of the control system and instrumentation

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Raw water pumping	5	2014	Like-for-like replacement of mechanical and electrical equipment
Coagulation and flocculation	1	1974	No major work; routine maintenance only
Clarifiers	1	2012	Like-for-like replacement of mechanical and electrical equipment
Rapid gravity filters	4	2003	Capacity uprating and new standby unit
Granular activated carbon	4	1959	Refurbishment of media and underdrains
Chlorine contact tank	1	1988	Refurbishment of media and underdrains
Treated water pumping	1	2010	Refurbishment of media and underdrains

Several process stages have no standby unit, so planned maintenance has to be carried out at night in winter. Energy use at the site was 511 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.2 Caddon works

Caddon is a site for river abstraction and membrane filtration, built in 2004, with a maximum output of 41 megalitres per day. In 2025/26 it supplied an average of 26.6 megalitres per day and was out of service for 67 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Intake and screens	3	2023	No major work; routine maintenance only
Raw water storage lagoon	3	2017	Like-for-like replacement of mechanical and electrical equipment
Ultrafiltration membranes	1	2062	Like-for-like replacement of mechanical and electrical equipment
Ultraviolet disinfection	1	2060	No major work; routine maintenance only
Chlorine dosing	4	2055	No major work; routine maintenance only
Treated water pumping	5	2031	Capacity uprating and new standby unit

Raw water quality deteriorates after heavy rain, and the works must reduce output when turbidity is high. Energy use at the site was 585 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.3 Elm Valley works

Elm Valley is a site for groundwater from eleven boreholes, built in 1936, with a maximum output of 38 megalitres per day. In 2025/26 it supplied an average of 27.3 megalitres per day and was out of service for 21 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Borehole pumps	3	1974	Structural repairs and new access walkways
Aeration tower	5	1936	Structural repairs and new access walkways

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Pressure filters	4	1987	Like-for-like replacement of mechanical and electrical equipment
Chlorine dosing	1	1944	Structural repairs and new access walkways
Contact main	4	1938	Structural repairs and new access walkways
Treated water pumping	5	1942	Structural repairs and new access walkways

Several process stages have no standby unit, so planned maintenance has to be carried out at night in winter. Energy use at the site was 372 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.4 Netherfold works

Netherfold is a site for impounding reservoir and conventional treatment, built in 1971, with a maximum output of 52 megalitres per day. In 2025/26 it supplied an average of 31.8 megalitres per day and was out of service for 48 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Draw-off tower	5	1979	Refurbishment of media and underdrains
Microstrainers	1	1996	No major work; routine maintenance only
Dissolved air flotation	2	1984	Capacity uprating and new standby unit
Rapid gravity filters	2	2020	Capacity uprating and new standby unit
Manganese contactors	2	1989	Structural repairs and new access walkways
Chlorine contact tank	1	2013	Capacity uprating and new standby unit
Clear water tank	3	2006	No major work; routine maintenance only

The site has a single power supply, and the standby generator can run only the treated water pumps. Energy use at the site was 453 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.5 Stavely Cross works

Stavely Cross is a site for spring sources with ultraviolet treatment, built in 1989, with a maximum output of 9 megalitres per day. In 2025/26 it supplied an average of 6.7 megalitres per day and was out of service for 10 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Spring collection chambers	2	2009	Refurbishment of media and underdrains
Cartridge filters	5	2022	No major work; routine maintenance only
Ultraviolet reactors	3	1989	Like-for-like replacement of mechanical and electrical equipment
Chlorine dosing	3	2022	Replacement of dosing pumps and chemical storage
Treated water storage	2	2032	Capacity uprating and new standby unit

Several process stages have no standby unit, so planned maintenance has to be carried out at night in winter. Energy use at the site was 548 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.6 Gorsemere works

Gorsemere is a site for lake abstraction and slow sand filtration, built in 1902, with a maximum output of 27 megalitres per day. In 2025/26 it supplied an average of 19.5 megalitres per day and was out of service for 68 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Lake intake	3	1909	No major work; routine maintenance only
Roughing filters	1	1928	No major work; routine maintenance only
Slow sand filters	1	1945	Capacity uprating and new standby unit
Chlorine contact tank	2	1912	Structural repairs and new access walkways
Treated water pumping	2	1960	Capacity uprating and new standby unit

Raw water quality deteriorates after heavy rain, and the works must reduce output when turbidity is high. Energy use at the site was 238 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.7 Larkhollow works

Larkhollow is a site for groundwater from four boreholes with nitrate removal, built in 1998, with a maximum output of 14 megalitres per day. In 2025/26 it supplied an average of 9.1 megalitres per day and was out of service for 37 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Borehole pumps	4	2006	Refurbishment of media and underdrains
Ion exchange nitrate removal	3	2052	Capacity uprating and new standby unit
Blending tank	2	2056	Replacement of dosing pumps and chemical storage
Chlorine dosing	4	2015	Like-for-like replacement of mechanical and electrical equipment
Treated water pumping	3	2008	Refurbishment of media and underdrains

Several process stages have no standby unit, so planned maintenance has to be carried out at night in winter. Energy use at the site was 352 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.8 Marrowby works

Marrowby is a site for river abstraction and conventional treatment, built in 1966, with a maximum output of 21 megalitres per day. In 2025/26 it supplied an average of 15.1 megalitres per day and was out of service for 4 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Intake and screens	5	1997	Replacement of dosing pumps and chemical storage

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Coagulation and flocculation	2	2006	Like-for-like replacement of mechanical and electrical equipment
Clarifiers	3	1994	Replacement of dosing pumps and chemical storage
Rapid gravity filters	2	1979	Replacement of the control system and instrumentation
Chlorine contact tank	5	2018	Replacement of dosing pumps and chemical storage
Sludge treatment	3	2015	Replacement of dosing pumps and chemical storage

The site has a single power supply, and the standby generator can run only the treated water pumps. Energy use at the site was 272 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

4.9 Tolland Edge works

Tolland Edge is a site for groundwater from two boreholes, built in 1979, with a maximum output of 10 megalitres per day. In 2025/26 it supplied an average of 6.0 megalitres per day and was out of service for 68 hours, mostly for planned maintenance.

Process stage	Condition grade	Last major renewal	Planned 2027–2036
Borehole pumps	4	2030	Replacement of dosing pumps and chemical storage
Pressure filters	3	2019	Structural repairs and new access walkways
Chlorine dosing	4	2005	Replacement of dosing pumps and chemical storage
Treated water storage	5	2006	Refurbishment of media and underdrains

Raw water quality deteriorates after heavy rain, and the works must reduce output when turbidity is high. Energy use at the site was 182 kWh per megalitre treated, and the plan includes a review of pump efficiency at the next major renewal.

5 Supply zone profiles

This chapter describes each of the twenty-four supply zones in the same order as the regional supply map. For each zone it gives an asset register summary, the condition and performance of the assets, the main risks and the interventions planned for 2027–2036. Intervention references (AMP-zone-number) are used in the capital programme in Chapter 6.

5.1 Ardley Fen supply zone

The Ardley Fen zone serves approximately 77,200 people in 31,530 properties, centred on a former mill town of Ardley Fen and the surrounding parishes. It is supplied from the Stavely Cross springs and ultraviolet plant and has 1 service reservoir with a combined storage of 19.0 megalitres, equivalent to about 2 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	111.0	km	11	2
Distribution mains	460.3	km	34	3
Service reservoirs	1	no.	73	2

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Pumping stations	6	no.	36	5
Pressure-reducing valves	14	no.	32	1
Hydrants	2,153	no.	20	1
Customer meters	23,654	no.	72	5
Telemetry outstations	39	no.	64	3

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
ARD-01	2,510	10.8	169	55.9
ARD-02	550	12.4	184	33.5
ARD-03	490	8.9	177	49.5
ARD-04	1,130	8.2	69	59.0
ARD-05	2,390	9.5	72	60.9
ARD-06	1,850	1.7	104	53.1

Night flows have fallen by 14 per cent since the district meters were renewed, but the zone still has three areas with pressures above 70 m.

Condition and performance

Ground movement in the clay soils causes a seasonal peak in bursts in late summer. Over the last three years the zone averaged 31.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 4.09 megalitres per day, and the average customer lost 41.5 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	293	0
Free chlorine residual	86	0
Iron	197	2
Manganese	197	0
Lead	384	0
Coliform bacteria	349	1

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The main risk is the single feed from the source works; an outage of more than eighteen hours would exhaust storage. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,010 new homes in the zone up to 2036. On current assumptions they will add 11.6 per cent to peak demand, and the existing network can absorb this growth until about 2030, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	2.6	86	772
2028/29	0.8	5	228
2029/30	4.0	4	627
2030/31	1.8	62	779
2031/32	5.4	126	85
2032/33	5.5	71	356
2033/34	0.8	22	267
2034/35	0.5	18	145
2035/36	3.2	219	828
2036/37	0.5	160	820

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-01-01	Upgrade 18 telemetry outstations to cellular communications	2035	360
AMP-01-02	Install 13 acoustic leak loggers on strategic mains	2028	50
AMP-01-03	Refurbish Ardley Fen No. 1 service reservoir (roof membrane and internal coating)	2029	1,320
AMP-01-04	Renew booster pumps and controls at Market Place pumping station	2030	480
AMP-01-05	New 4.6 Ml storage tank at Market Place	2034	6,690
AMP-01-06	Pressure management in 2 district metered areas	2035	520
	Total for Ardley Fen		**9,420**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

Planning consent is needed for the new reservoir; pre-application discussions have started. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.2 Brackwater supply zone

The Brackwater zone serves approximately 57,200 people in 24,660 properties, centred on a commuter town of Brackwater and the surrounding parishes. It is supplied from the Stavely Cross springs and ultraviolet plant and has 3 service reservoirs with a combined storage of 7.8 megalitres, equivalent to about 1 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	10.9	km	21	3
Distribution mains	95.9	km	39	4
Service reservoirs	3	no.	55	1

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Pumping stations	7	no.	37	5
Pressure-reducing valves	62	no.	8	2
Hydrants	2,154	no.	67	5
Customer meters	20,300	no.	31	5
Telemetry outstations	37	no.	65	5

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
BRA-01	400	7.7	44	67.5
BRA-02	1,120	6.0	53	30.0
BRA-03	1,550	12.5	110	46.6
BRA-04	2,280	7.4	130	47.5
BRA-05	540	1.9	174	46.7
BRA-06	3,110	11.3	152	60.3
BRA-07	1,600	7.2	50	37.3
BRA-08	3,080	4.6	89	62.3

Most of the zone is fed by gravity from the service reservoirs; only 13 per cent of properties depend on a booster pumping station.

Condition and performance

Mains condition is generally fair, but the cast-iron mains laid between 1890 and 1935 account for most bursts. Over the last three years the zone averaged 26.5 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 0.85 megalitres per day, and the average customer lost 38.1 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	89	0
Free chlorine residual	275	0
Iron	114	0
Manganese	343	0
Lead	163	0
Coliform bacteria	308	2

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The service reservoirs are close to the end of their structural life and two have known roof defects. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,030 new homes in the zone up to 2036. On current assumptions they will add 10.6 per cent to peak demand, and the existing network can absorb this growth until about 2033, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	5.2	23	340
2028/29	3.0	128	589
2029/30	6.0	17	207
2030/31	2.4	164	56
2031/32	4.4	191	807
2032/33	0.7	195	205
2033/34	4.3	83	366
2034/35	2.5	179	196
2035/36	2.3	110	401
2036/37	5.3	196	160

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-02-01	New 1.7 Ml storage tank at Copse Road	2035	2,260
AMP-02-02	Replace 4.7 km of trunk main along Long Furlong	2032	4,790
AMP-02-03	Upgrade 6 telemetry outstations to cellular communications	2030	80
AMP-02-04	Rehabilitate 2.3 km of cast-iron distribution mains	2028	1,070
AMP-02-05	Refurbish Brackwater No. 3 service reservoir (roof membrane and internal coating)	2030	1,730
	Total for Brackwater		**9,930**

The option of doing nothing was costed in each case and used as the baseline for the benefit assessment.

Customer contact will be handled by the regional engagement team, with letters sent four weeks before work starts. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.3 Caddon Heath supply zone

The Caddon Heath zone serves approximately 53,300 people in 23,550 properties, centred on a former mill town of Caddon Heath and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 2 service reservoirs with a combined storage of 27.5 megalitres, equivalent to about 4 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	71.3	km	26	5
Distribution mains	274.5	km	22	5
Service reservoirs	2	no.	44	5
Pumping stations	1	no.	52	1

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Pressure-reducing valves	9	no.	14	1
Hydrants	535	no.	48	5
Customer meters	19,329	no.	10	4
Telemetry outstations	60	no.	22	1

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
CAD-01	2,140	14.8	198	32.1
CAD-02	3,230	8.5	46	60.1
CAD-03	870	14.4	88	64.5
CAD-04	2,910	9.0	142	55.3
CAD-05	2,700	2.2	41	36.3
CAD-06	2,410	10.2	114	53.6
CAD-07	2,320	4.6	169	44.3
CAD-08	3,210	8.3	161	66.2
CAD-09	2,360	11.1	146	30.1
CAD-10	1,560	3.0	43	48.4
CAD-11	910	3.6	153	66.0
CAD-12	430	13.6	198	61.2
CAD-13	1,900	6.7	41	44.1

The hydraulic model was recalibrated in 2025 using 49 pressure loggers; it shows that the trunk main along School Lane runs close to its capacity on peak days.

Condition and performance

Ground movement in the clay soils causes a seasonal peak in bursts in late summer. Over the last three years the zone averaged 20.4 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 1.37 megalitres per day, and the average customer lost 5.1 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	46	0
Free chlorine residual	116	0
Iron	346	0
Manganese	114	1
Lead	279	2
Coliform bacteria	192	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Rising peak summer demand from new housing is reducing the margin between demand and pumping capacity. The access track to the Caddon Heath reservoir floods in winter, which delays emergency repairs. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 150 new homes in the zone up to 2036. On current assumptions they will add 13.3 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	5.8	39	879
2028/29	4.9	26	812
2029/30	2.9	154	356
2030/31	0.9	163	444
2031/32	6.0	149	294
2032/33	3.1	67	689
2033/34	2.8	4	752
2034/35	1.1	158	884
2035/36	4.6	125	574
2036/37	4.8	85	864

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-03-01	Install 20 acoustic leak loggers on strategic mains	2027	70
AMP-03-02	Replace 343 lead communication pipes in the School Lane area	2027	1,100
AMP-03-03	Pressure management in 8 district metered areas	2035	1,260
AMP-03-04	Rehabilitate 2.3 km of cast-iron distribution mains	2032	1,200
AMP-03-05	Refurbish Caddon Heath No. 2 service reservoir (roof membrane and internal coating)	2035	1,150
	Total for Caddon Heath		**4,780**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

Works in this zone are coordinated with the highway authority’s resurfacing programme to avoid digging up new road surfaces. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.4 Dunholt supply zone

The Dunholt zone serves approximately 75,900 people in 30,610 properties, centred on two small towns of Dunholt and the surrounding parishes. It is supplied from the Caddon river intake and treatment works and has 5 service reservoirs with a combined storage of 30.4 megalitres, equivalent to about 3 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	96.2	km	42	3
Distribution mains	587.9	km	16	3
Service reservoirs	5	no.	38	3
Pumping stations	3	no.	28	2
Pressure-reducing valves	19	no.	59	1
Hydrants	308	no.	76	5
Customer meters	28,991	no.	34	5
Telemetry outstations	50	no.	40	1

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
DUN-01	2,940	15.0	135	54.9
DUN-02	1,180	8.0	143	37.0
DUN-03	2,590	9.1	126	29.2
DUN-04	1,800	3.6	114	67.6
DUN-05	1,490	13.2	133	41.9
DUN-06	3,200	13.6	81	62.6

The critical point for pressure is at Kiln Street, where the service level of 20 m is met only when both booster pumps are available.

Condition and performance

Most of the network was laid after 1960 in ductile iron and PVC, and its condition is good for its age. Over the last three years the zone averaged 28.1 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 4.40 megalitres per day, and the average customer lost 36.6 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	69	3
Free chlorine residual	394	0
Iron	59	0
Manganese	107	0
Lead	301	2
Coliform bacteria	61	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Rising peak summer demand from new housing is reducing the margin between demand and pumping capacity. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,400 new homes in the zone up to 2036. On current assumptions they will add 13.3 per cent to peak demand, and the existing network can absorb this growth until about 2030, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	5.3	71	122
2028/29	1.4	47	151
2029/30	2.6	21	448
2030/31	3.6	174	794
2031/32	5.2	110	126
2032/33	4.6	41	436
2033/34	1.6	49	630
2034/35	2.5	132	537
2035/36	1.6	152	803
2036/37	4.2	197	309

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-04-01	Replace 5.2 km of trunk main along School Lane	2032	7,790
AMP-04-02	Upgrade 18 telemetry outstations to cellular communications	2032	220
AMP-04-03	Refurbish Dunholt No. 2 service reservoir (roof membrane and internal coating)	2033	1,020
AMP-04-04	New 4.4 Ml storage tank at Kiln Street	2035	7,350
AMP-04-05	Install 42 acoustic leak loggers on strategic mains	2031	130
	Total for Dunholt		**16,510**

Lining rather than replacing the mains was rejected because of the number of service connections, which would each need to be re-made.

Works in this zone are coordinated with the highway authority’s resurfacing programme to avoid digging up new road surfaces. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.5 Elmscar supply zone

The Elmscar zone serves approximately 32,100 people in 13,270 properties, centred on a coastal resort of Elmscar and the surrounding parishes. It is supplied from the Stavely Cross springs and ultraviolet plant and has 4 service reservoirs with a combined storage of 26.9 megalitres, equivalent to about 6 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	148.6	km	67	5
Distribution mains	530.0	km	30	4
Service reservoirs	4	no.	18	4
Pumping stations	7	no.	70	1
Pressure-reducing valves	27	no.	79	4
Hydrants	121	no.	68	1
Customer meters	9,921	no.	26	3
Telemetry outstations	58	no.	64	1

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
ELM-01	1,580	6.1	151	67.4
ELM-02	2,920	1.8	159	35.8
ELM-03	3,000	2.7	129	59.8
ELM-04	1,490	4.8	106	45.9
ELM-05	2,260	14.6	151	55.8
ELM-06	3,320	1.2	48	28.4
ELM-07	3,110	11.6	160	63.7
ELM-08	1,820	11.3	168	31.3

Night flows have fallen by 16 per cent since the district meters were renewed, but the zone still has three areas with pressures above 70 m.

Condition and performance

Most of the network was laid after 1960 in ductile iron and PVC, and its condition is good for its age. Over the last three years the zone averaged 25.6 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 1.81 megalitres per day, and the average customer lost 38.2 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	121	0
Free chlorine residual	333	0
Iron	326	0
Manganese	366	2
Lead	280	0
Coliform bacteria	235	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Rising peak summer demand from new housing is reducing the margin between demand and pumping capacity. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,270 new homes in the zone up to 2036. On current assumptions they will add 9.7 per cent to peak demand, and the existing network can absorb this growth until about 2028, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.5	116	616
2028/29	3.3	151	732
2029/30	1.3	127	482
2030/31	0.8	169	571
2031/32	0.9	204	244
2032/33	5.9	6	33
2033/34	0.0	161	840
2034/35	4.9	76	831
2035/36	4.4	152	163
2036/37	5.7	139	382

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-05-01	New 3.5 Ml storage tank at Ferry Road	2032	4,840
AMP-05-02	Install 36 acoustic leak loggers on strategic mains	2028	90
AMP-05-03	Refurbish Elmscar No. 2 service reservoir (roof membrane and internal coating)	2030	1,220
AMP-05-04	Replace 5.1 km of trunk main along Station Road	2034	6,900
	Total for Elmscar		**13,050**

Lining rather than replacing the mains was rejected because of the number of service connections, which would each need to be re-made.

The interventions depend on the power supply upgrade that the distribution network operator has scheduled. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.6 Fernlow supply zone

The Fernlow zone serves approximately 86,800 people in 37,330 properties, centred on a former mill town of Fernlow and the surrounding parishes. It is supplied from the Caddon river intake and treatment works and has 6 service reservoirs with a combined storage of 4.2 megalitres, equivalent to about 0 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	142.2	km	47	4

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Distribution mains	375.0	km	17	2
Service reservoirs	6	no.	31	2
Pumping stations	9	no.	80	2
Pressure-reducing valves	13	no.	17	1
Hydrants	1,937	no.	30	1
Customer meters	25,946	no.	24	1
Telemetry outstations	61	no.	32	5

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
FER-01	720	3.8	111	30.9
FER-02	2,210	3.3	41	38.3
FER-03	3,090	12.5	47	65.5
FER-04	1,580	9.5	58	39.8
FER-05	1,810	2.3	68	60.5
FER-06	790	10.9	63	53.7
FER-07	1,940	6.6	121	63.2
FER-08	1,570	7.0	151	34.7
FER-09	930	4.1	169	56.5
FER-10	2,690	13.0	152	44.9
FER-11	3,300	7.2	41	39.9

Pressures range from 19 m at the top of Chapel Hill to 83 m in the lower part of the zone, so the pressure-reducing valves are set to keep night pressures below 45 m.

Condition and performance

The zone has a high share of small-diameter galvanised steel mains, which corrode from the inside. Over the last three years the zone averaged 24.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 2.34 megalitres per day, and the average customer lost 21.6 minutes of supply per year to unplanned interruptions. The 1912 brick culvert carrying the overflow from the upper reservoir was surveyed by remote camera and found sound.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	158	0
Free chlorine residual	139	0
Iron	174	3
Manganese	173	3
Lead	263	0
Coliform bacteria	375	0

Complaints about low pressure have fallen since the booster pumps were replaced, but remain high in the upper part of the zone. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Rising peak summer demand from new housing is reducing the margin between demand and pumping capacity. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,180 new homes in the zone up to 2036. On current assumptions they will add 10.3 per cent to peak demand, and the existing network can absorb this growth until about 2033, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.6	3	740
2028/29	4.6	91	364
2029/30	5.0	28	699
2030/31	2.2	111	592
2031/32	3.6	113	363
2032/33	4.3	47	145
2033/34	1.4	115	234
2034/35	0.1	128	725
2035/36	4.3	41	616
2036/37	3.6	2	880

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-06-01	Refurbish Fernlow No. 2 service reservoir (roof membrane and internal coating)	2032	1,780
AMP-06-02	Pressure management in 7 district metered areas	2034	880
AMP-06-03	New 1.9 Ml storage tank at Chapel Hill	2027	2,920
AMP-06-04	Renew booster pumps and controls at Chapel Hill pumping station	2035	800
AMP-06-05	Replace 2.3 km of trunk main along Gallows Hill	2027	2,290
AMP-06-06	Replace 340 lead communication pipes in the Gallows Hill area	2031	1,020
	Total for Fernlow		**9,690**

Pressure management was preferred to additional mains renewal because it reduces bursts across the whole zone, not only on the renewed lengths.

Works in this zone are coordinated with the highway authority’s resurfacing programme to avoid digging up new road surfaces. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.7 Gorsemere supply zone

The Gorsemere zone serves approximately 48,000 people in 21,520 properties, centred on two small towns of Gorsemere and the surrounding parishes. It is supplied from the Netherfold impounding reservoir and has 2 service reservoirs with a combined storage of 11.4 megalitres, equivalent to about 2 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	38.9	km	19	3
Distribution mains	62.7	km	35	5
Service reservoirs	2	no.	27	2
Pumping stations	7	no.	49	2
Pressure-reducing valves	14	no.	50	2
Hydrants	1,075	no.	14	2
Customer meters	13,448	no.	10	5
Telemetry outstations	21	no.	38	3

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
GOR-01	460	1.8	73	61.6
GOR-02	770	8.4	47	63.7
GOR-03	3,010	6.4	73	53.8
GOR-04	3,420	3.2	93	46.8
GOR-05	2,870	5.0	155	31.9
GOR-06	2,780	6.4	57	67.3
GOR-07	1,880	11.6	196	34.0
GOR-08	2,860	2.6	110	41.1

The hydraulic model was recalibrated in 2023 using 75 pressure loggers; it shows that the trunk main along Quarry Lane runs close to its capacity on peak days.

Condition and performance

The zone has a high share of small-diameter galvanised steel mains, which corrode from the inside. Over the last three years the zone averaged 31.4 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 3.54 megalitres per day, and the average customer lost 31.5 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	329	0
Free chlorine residual	188	0
Iron	304	3
Manganese	117	0
Lead	52	0

Parameter (2025)	Samples	Samples above standard
Coliform bacteria	258	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Rising peak summer demand from new housing is reducing the margin between demand and pumping capacity. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 890 new homes in the zone up to 2036. On current assumptions they will add 8.1 per cent to peak demand, and the existing network can absorb this growth until about 2029, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	2.6	154	568
2028/29	0.8	44	13
2029/30	1.0	86	754
2030/31	0.2	23	487
2031/32	1.3	132	838
2032/33	2.2	187	838
2033/34	1.9	46	236
2034/35	0.5	174	714
2035/36	2.0	112	406
2036/37	0.0	14	6

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-07-01	Refurbish Gorsemere No. 1 service reservoir (roof membrane and internal coating)	2030	2,990
AMP-07-02	Renew booster pumps and controls at Sandpit Lane pumping station	2035	490
AMP-07-03	Rehabilitate 10.8 km of cast-iron distribution mains	2027	7,210
AMP-07-04	Replace 180 lead communication pipes in the Quarry Lane area	2035	500
	Total for Gorsemere		**11,190**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

Planning consent is needed for the new reservoir; pre-application discussions have started. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.8 Hatherby supply zone

The Hatherby zone serves approximately 65,800 people in 27,840 properties, centred on an expanding new settlement of Hatherby and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 1 service reservoir with a combined storage of 2.6 megalitres, equivalent to about 0 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	73.1	km	36	3
Distribution mains	167.6	km	37	2
Service reservoirs	1	no.	41	3
Pumping stations	2	no.	49	4
Pressure-reducing valves	28	no.	25	5
Hydrants	1,263	no.	50	1
Customer meters	18,186	no.	32	5
Telemetry outstations	13	no.	26	4

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
HAT-01	3,160	2.8	49	29.1
HAT-02	790	14.0	117	55.7
HAT-03	740	13.3	87	47.8
HAT-04	1,720	13.3	111	45.6
HAT-05	710	6.8	50	48.4
HAT-06	1,360	1.2	130	61.2

Pressures range from 24 m at the top of Market Place to 80 m in the lower part of the zone, so the pressure-reducing valves are set to keep night pressures below 45 m.

Condition and performance

Most of the network was laid after 1960 in ductile iron and PVC, and its condition is good for its age. Over the last three years the zone averaged 8.5 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 1.88 megalitres per day, and the average customer lost 4.0 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	336	0
Free chlorine residual	271	0
Iron	158	0
Manganese	176	0
Lead	190	0
Coliform bacteria	368	0

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The main risk is the single feed from the source works; an outage of more than eighteen hours would exhaust storage. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,270 new homes in the zone up to 2036. On current assumptions they will add 10.0 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	2.9	122	9
2028/29	5.6	56	835
2029/30	1.9	13	42
2030/31	5.9	19	590
2031/32	4.5	162	156
2032/33	4.5	68	76
2033/34	2.9	9	151
2034/35	2.1	80	248
2035/36	5.4	216	629
2036/37	0.7	103	513

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-08-01	New 2.2 Ml storage tank at Market Place	2029	3,330
AMP-08-02	Replace 156 lead communication pipes in the Chapel Hill area	2029	420
AMP-08-03	Replace 6.7 km of trunk main along Chapel Hill	2029	9,160
AMP-08-04	Refurbish Hatherby No. 2 service reservoir (roof membrane and internal coating)	2030	1,640
AMP-08-05	Pressure management in 5 district metered areas	2032	860
AMP-08-06	Renew booster pumps and controls at Market Place pumping station	2035	860
	Total for Hatherby		**16,270**

A second feed into the zone was considered, but its cost was not justified by the reduction in risk; additional storage was chosen instead.

The programme assumes that the land purchase for the new pumping station completes in the first year. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.9 Ivelstone supply zone

The Ivelstone zone serves approximately 56,900 people in 23,150 properties, centred on a former mill town of Ivelstone and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 3 service reservoirs with a combined storage of 14.4 megalitres, equivalent to about 2 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	35.7	km	13	1
Distribution mains	328.5	km	43	4
Service reservoirs	3	no.	21	2
Pumping stations	2	no.	48	1
Pressure-reducing valves	37	no.	78	5
Hydrants	1,702	no.	23	4
Customer meters	20,404	no.	24	5
Telemetry outstations	18	no.	29	5

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
IVE-01	3,370	1.4	198	54.7
IVE-02	1,880	3.7	190	56.5
IVE-03	1,360	5.2	52	51.1
IVE-04	2,030	6.5	174	41.1
IVE-05	500	8.5	62	34.2
IVE-06	2,710	13.4	191	61.8
IVE-07	1,630	5.3	123	39.6

The hydraulic model was recalibrated in 2024 using 45 pressure loggers; it shows that the trunk main along Beacon Rise runs close to its capacity on peak days.

Condition and performance

Mains condition is generally fair, but the cast-iron mains laid between 1890 and 1935 account for most bursts. Over the last three years the zone averaged 16.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 1.68 megalitres per day, and the average customer lost 5.5 minutes of supply per year to unplanned interruptions. A further 2.3 km of cast-iron main beneath Harbour Street will be relined during the quay works, under a joint traffic management plan with the highway authority.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	20	0
Free chlorine residual	172	0
Iron	41	0
Manganese	340	0
Lead	104	0

Parameter (2025)	Samples	Samples above standard
Coliform bacteria	31	0

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Low pressure at the top of the zone is caused by an undersized main rather than by the pumps. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,960 new homes in the zone up to 2036. On current assumptions they will add 13.2 per cent to peak demand, and the existing network can absorb this growth until about 2028, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	3.0	178	44
2028/29	1.6	173	378
2029/30	0.8	87	197
2030/31	2.0	106	777
2031/32	2.7	116	43
2032/33	3.2	16	623
2033/34	3.5	119	78
2034/35	0.7	158	128
2035/36	0.1	208	118
2036/37	5.3	179	137

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-09-01	Replace 1.5 km of trunk main along Beacon Rise	2028	1,850
AMP-09-02	Refurbish Ivelstone No. 2 service reservoir (roof membrane and internal coating)	2027	2,530
AMP-09-03	New 1.8 Ml storage tank at Copse Road	2027	2,910
AMP-09-04	Install 19 acoustic leak loggers on strategic mains	2035	70
	Total for Ivelstone		**7,360**

For the reservoir defects, the options were refurbishment, replacement on the same site and a new tank on a higher site; refurbishment had the lowest whole-life cost.

The programme assumes that the land purchase for the new pumping station completes in the first year. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.10 Kelmarsh Vale supply zone

The Kelmarsh Vale zone serves approximately 82,200 people in 33,150 properties, centred on an expanding new settlement of Kelmarsh Vale and the surrounding parishes. It is supplied from the Netherfold impounding reservoir

and has 1 service reservoir with a combined storage of 38.6 megalitres, equivalent to about 3 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	64.2	km	33	2
Distribution mains	63.8	km	69	4
Service reservoirs	1	no.	79	5
Pumping stations	2	no.	8	2
Pressure-reducing valves	3	no.	13	1
Hydrants	2,136	no.	11	5
Customer meters	23,653	no.	78	5
Telemetry outstations	62	no.	26	5

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
KEL-01	3,260	2.4	55	30.4
KEL-02	1,170	14.1	136	50.7
KEL-03	640	12.7	195	30.3
KEL-04	1,630	4.0	116	38.0
KEL-05	2,980	4.2	144	52.8
KEL-06	1,770	13.5	195	61.4

Most of the zone is fed by gravity from the service reservoirs; only 23 per cent of properties depend on a booster pumping station.

Condition and performance

The zone has a high share of small-diameter galvanised steel mains, which corrode from the inside. Over the last three years the zone averaged 20.6 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 3.45 megalitres per day, and the average customer lost 13.1 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	127	0
Free chlorine residual	274	0
Iron	289	0
Manganese	321	0
Lead	224	3
Coliform bacteria	342	0

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The service reservoirs are close to the end of their structural life and two have known roof defects. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,430 new homes in the zone up to 2036. On current assumptions they will add 15.1 per cent to peak demand, and the existing network can absorb this growth until about 2029, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	5.9	176	587
2028/29	4.9	37	845
2029/30	1.9	202	307
2030/31	0.6	135	818
2031/32	1.8	79	711
2032/33	1.5	171	264
2033/34	6.0	10	303
2034/35	5.4	63	481
2035/36	1.3	6	438
2036/37	6.0	161	251

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-10-01	Renew booster pumps and controls at Long Furlong pumping station	2030	970
AMP-10-02	Upgrade 21 telemetry outstations to cellular communications	2028	350
AMP-10-03	Install 24 acoustic leak loggers on strategic mains	2029	70
AMP-10-04	New 2.3 Ml storage tank at Long Furlong	2032	4,170
AMP-10-05	Replace 4.5 km of trunk main along Station Road	2029	5,500
AMP-10-06	Refurbish Kelmarsh Vale No. 1 service reservoir (roof membrane and internal coating)	2033	1,640
	Total for Kelmarsh Vale		**12,700**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

Planning consent is needed for the new reservoir; pre-application discussions have started. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.11 Larkhollow supply zone

The Larkhollow zone serves approximately 24,600 people in 10,560 properties, centred on the market town of Larkhollow and the surrounding parishes. It is supplied from the Stavely Cross springs and ultraviolet plant and has 4 service reservoirs with a combined storage of 21.5 megalitres, equivalent to about 6 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	31.5	km	27	4
Distribution mains	178.6	km	32	3
Service reservoirs	4	no.	70	1
Pumping stations	9	no.	68	2
Pressure-reducing valves	22	no.	85	4
Hydrants	1,918	no.	87	2
Customer meters	8,567	no.	12	2
Telemetry outstations	57	no.	80	4

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
LAR-01	3,370	9.8	129	42.2
LAR-02	440	13.6	112	61.9
LAR-03	1,250	8.1	178	59.7
LAR-04	1,060	4.8	114	56.4
LAR-05	1,250	8.0	67	49.3
LAR-06	1,000	13.2	99	51.8
LAR-07	530	1.4	59	52.3
LAR-08	720	5.4	77	54.9
LAR-09	2,480	13.5	162	42.8
LAR-10	1,490	4.2	53	66.7
LAR-11	3,200	14.9	72	29.8

Most of the zone is fed by gravity from the service reservoirs; only 4 per cent of properties depend on a booster pumping station.

Condition and performance

The asbestos-cement mains installed in the 1950s and 1960s are now the main source of failures. Over the last three years the zone averaged 31.4 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 7.70 megalitres per day, and the average customer lost 14.3 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	280	0
Free chlorine residual	51	0
Iron	325	0
Manganese	242	0
Lead	269	3

Parameter (2025)	Samples	Samples above standard
Coliform bacteria	292	0

Customers in the zone said in the consultation that they would accept longer roadworks if the number of interruptions fell. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The main risk is the single feed from the source works; an outage of more than eighteen hours would exhaust storage. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 200 new homes in the zone up to 2036. On current assumptions they will add 5.3 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.6	150	786
2028/29	0.3	52	451
2029/30	4.0	47	435
2030/31	3.6	47	443
2031/32	3.8	62	792
2032/33	3.0	95	785
2033/34	0.6	175	656
2034/35	5.1	88	5
2035/36	3.3	108	885
2036/37	2.2	56	68

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-11-01	Upgrade 7 telemetry outstations to cellular communications	2032	90
AMP-11-02	Replace 201 lead communication pipes in the Ferry Road area	2034	610
AMP-11-03	Replace 4.5 km of trunk main along Ferry Road	2034	4,950
AMP-11-04	New 3.5 Ml storage tank at Tanners Row	2034	6,360
	Total for Larkhollow		**12,010**

Lining rather than replacing the mains was rejected because of the number of service connections, which would each need to be re-made.

The interventions depend on the power supply upgrade that the distribution network operator has scheduled. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.12 Marrowby supply zone

The Marrowby zone serves approximately 44,400 people in 20,130 properties, centred on a commuter town of Marrowby and the surrounding parishes. It is supplied from the Stavely Cross springs and ultraviolet plant and has 5 service reservoirs with a combined storage of 34.6 megalitres, equivalent to about 6 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	30.0	km	44	2
Distribution mains	121.4	km	26	5
Service reservoirs	5	no.	35	4
Pumping stations	9	no.	85	3
Pressure-reducing valves	39	no.	77	2
Hydrants	1,026	no.	83	2
Customer meters	18,832	no.	35	5
Telemetry outstations	55	no.	35	4

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
MAR-01	1,960	4.1	163	59.2
MAR-02	1,270	2.1	187	59.6
MAR-03	1,190	11.4	147	33.5
MAR-04	2,880	3.6	65	65.1
MAR-05	1,850	2.1	124	65.7
MAR-06	2,310	4.8	117	42.8
MAR-07	1,010	4.2	137	51.5
MAR-08	2,800	11.3	150	52.4

Most of the zone is fed by gravity from the service reservoirs; only 16 per cent of properties depend on a booster pumping station.

Condition and performance

Condition is mixed: the town centre has old cast-iron mains, while the outer estates have modern polyethylene pipes. Over the last three years the zone averaged 11.1 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 7.41 megalitres per day, and the average customer lost 43.4 minutes of supply per year to unplanned interruptions. Burst frequency doubled during the dry summer of 2025, when soil shrinkage was at its worst.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	25	0
Free chlorine residual	213	0
Iron	364	2
Manganese	259	0

Parameter (2025)	Samples	Samples above standard
Lead	406	1
Coliform bacteria	219	2

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The service reservoirs are close to the end of their structural life and two have known roof defects. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,170 new homes in the zone up to 2036. On current assumptions they will add 10.4 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	3.7	184	286
2028/29	0.1	217	44
2029/30	3.3	183	639
2030/31	2.8	181	660
2031/32	0.5	94	316
2032/33	3.1	60	719
2033/34	0.7	12	222
2034/35	0.0	10	874
2035/36	5.8	190	92
2036/37	0.1	40	674

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-12-01	Upgrade 18 telemetry outstations to cellular communications	2029	280
AMP-12-02	Pressure management in 8 district metered areas	2035	1,050
AMP-12-03	Refurbish Marrowby No. 2 service reservoir (roof membrane and internal coating)	2035	1,400
AMP-12-04	Replace 5.4 km of trunk main along Tanners Row	2030	5,100
	Total for Marrowby		**7,830**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

Works in this zone are coordinated with the highway authority’s resurfacing programme to avoid digging up new road surfaces. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.13 Netherfold supply zone

The Netherfold zone serves approximately 82,100 people in 37,230 properties, centred on an expanding new settlement of Netherfold and the surrounding parishes. It is supplied from the Netherfold impounding reservoir and has 5 service reservoirs with a combined storage of 17.2 megalitres, equivalent to about 1 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	32.9	km	73	1
Distribution mains	208.4	km	79	5
Service reservoirs	5	no.	18	1
Pumping stations	1	no.	60	2
Pressure-reducing valves	51	no.	34	4
Hydrants	673	no.	87	3
Customer meters	21,145	no.	22	3
Telemetry outstations	36	no.	60	2

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
NET-01	1,710	14.1	89	61.3
NET-02	1,840	7.1	148	34.5
NET-03	450	4.0	53	59.8
NET-04	2,660	3.4	121	33.1
NET-05	2,900	7.6	88	54.6
NET-06	2,760	1.5	185	33.0
NET-07	1,940	5.0	152	65.5
NET-08	1,280	5.2	140	49.8
NET-09	3,450	11.0	129	35.2
NET-10	1,730	6.9	109	42.5
NET-11	660	4.0	42	35.6
NET-12	2,160	2.6	145	59.1

Pressures range from 28 m at the top of High Causeway to 65 m in the lower part of the zone, so the pressure-reducing valves are set to keep night pressures below 45 m.

Condition and performance

Condition is mixed: the town centre has old cast-iron mains, while the outer estates have modern polyethylene pipes. Over the last three years the zone averaged 19.7 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 4.90 megalitres per day, and the average customer lost 6.6 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	325	1
Free chlorine residual	307	0
Iron	137	1
Manganese	325	1
Lead	173	0
Coliform bacteria	12	2

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Low pressure at the top of the zone is caused by an undersized main rather than by the pumps. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,410 new homes in the zone up to 2036. On current assumptions they will add 8.4 per cent to peak demand, and the existing network can absorb this growth until about 2030, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	4.2	158	596
2028/29	0.7	154	320
2029/30	4.3	158	605
2030/31	1.1	32	583
2031/32	5.5	3	351
2032/33	3.1	193	478
2033/34	3.8	67	62
2034/35	0.3	94	275
2035/36	2.3	38	831
2036/37	4.8	212	760

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-13-01	Rehabilitate 12.8 km of cast-iron distribution mains	2035	8,730
AMP-13-02	Upgrade 21 telemetry outstations to cellular communications	2028	270
AMP-13-03	Pressure management in 4 district metered areas	2033	990
AMP-13-04	New 4.3 Ml storage tank at High Causeway	2029	7,180
AMP-13-05	Refurbish Netherfold No. 1 service reservoir (roof membrane and internal coating)	2033	1,570
AMP-13-06	Renew booster pumps and controls at High Causeway pumping station	2030	890

Ref.	Intervention	Start	Cost (£k)
	Total for Netherfold		**19,630**

Pressure management was preferred to additional mains renewal because it reduces bursts across the whole zone, not only on the renewed lengths.

The programme assumes that the land purchase for the new pumping station completes in the first year. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.14 Pellow Bay supply zone

The Pellow Bay zone serves approximately 54,700 people in 24,600 properties, centred on an expanding new settlement of Pellow Bay and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 2 service reservoirs with a combined storage of 16.3 megalitres, equivalent to about 2 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	61.4	km	21	5
Distribution mains	390.0	km	82	1
Service reservoirs	2	no.	14	3
Pumping stations	6	no.	32	2
Pressure-reducing valves	51	no.	28	1
Hydrants	420	no.	30	4
Customer meters	19,817	no.	49	2
Telemetry outstations	18	no.	52	1

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
PEL-01	2,210	13.3	50	62.5
PEL-02	1,190	2.3	148	57.9
PEL-03	3,170	7.4	196	59.4
PEL-04	3,210	12.9	102	34.6
PEL-05	3,210	5.5	163	45.5
PEL-06	2,200	11.0	59	41.0
PEL-07	340	13.1	66	58.6
PEL-08	2,420	13.7	128	38.9
PEL-09	2,020	4.4	176	37.6
PEL-10	990	4.0	63	30.6
PEL-11	1,230	13.2	140	41.4
PEL-12	1,810	9.5	194	37.1

Most of the zone is fed by gravity from the service reservoirs; only 14 per cent of properties depend on a booster pumping station.

Condition and performance

Most of the network was laid after 1960 in ductile iron and PVC, and its condition is good for its age. Over the last three years the zone averaged 18.6 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 2.61 megalitres per day, and the average customer lost 33.8 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	405	0
Free chlorine residual	63	0
Iron	265	0
Manganese	155	2
Lead	350	3
Coliform bacteria	320	0

Complaints about low pressure have fallen since the booster pumps were replaced, but remain high in the upper part of the zone. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Low pressure at the top of the zone is caused by an undersized main rather than by the pumps. The raw water intake screens at Pellow Bay are inspected quarterly and after every storm warning issued for the estuary. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 490 new homes in the zone up to 2036. On current assumptions they will add 15.5 per cent to peak demand, and the existing network can absorb this growth until about 2028, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	2.9	10	309
2028/29	5.6	150	740
2029/30	2.1	179	698
2030/31	5.9	173	312
2031/32	1.5	207	670
2032/33	4.0	75	402
2033/34	3.7	104	257
2034/35	4.7	34	341
2035/36	0.5	119	877
2036/37	0.2	93	148

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-14-01	Replace 6.6 km of trunk main along Weavers Close	2030	6,190

Ref.	Intervention	Start	Cost (£k)
AMP-14-02	Refurbish Pellow Bay No. 2 service reservoir (roof membrane and internal coating)	2034	1,250
AMP-14-03	Rehabilitate 12.8 km of cast-iron distribution mains	2035	7,810
AMP-14-04	Pressure management in 3 district metered areas	2034	400
	Total for Pellow Bay		**15,650**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

Planning consent is needed for the new reservoir; pre-application discussions have started. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.15 Quarrendon supply zone

The Quarrendon zone serves approximately 9,200 people in 3,870 properties, centred on the market town of Quarrendon and the surrounding parishes. It is supplied from the Caddon river intake and treatment works and has 4 service reservoirs with a combined storage of 4.5 megalitres, equivalent to about 4 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	107.4	km	25	2
Distribution mains	419.2	km	79	1
Service reservoirs	4	no.	33	3
Pumping stations	9	no.	50	1
Pressure-reducing valves	30	no.	60	2
Hydrants	402	no.	27	2
Customer meters	3,350	no.	13	4
Telemetry outstations	44	no.	20	4

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
QUA-01	2,200	10.8	92	59.1
QUA-02	2,670	8.5	145	36.4
QUA-03	850	12.8	117	58.1
QUA-04	740	2.6	152	61.7
QUA-05	840	10.1	96	33.1
QUA-06	2,020	8.9	120	46.0
QUA-07	3,000	12.3	109	33.1
QUA-08	2,900	9.1	140	28.2
QUA-09	1,540	10.1	200	56.2
QUA-10	370	3.1	165	43.7
QUA-11	2,220	8.2	109	39.3

Night flows have fallen by 23 per cent since the district meters were renewed, but the zone still has three areas with pressures above 70 m.

Condition and performance

Most of the network was laid after 1960 in ductile iron and PVC, and its condition is good for its age. Over the last three years the zone averaged 10.3 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 0.95 megalitres per day, and the average customer lost 19.4 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	251	0
Free chlorine residual	199	1
Iron	387	0
Manganese	141	0
Lead	82	0
Coliform bacteria	151	0

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Telemetry coverage is incomplete, so some reservoir levels are read only during site visits. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,190 new homes in the zone up to 2036. On current assumptions they will add 15.1 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	3.1	57	194
2028/29	5.4	74	496
2029/30	4.9	2	30
2030/31	2.8	159	593
2031/32	5.2	13	277
2032/33	0.5	120	468
2033/34	5.0	215	582
2034/35	5.0	206	843
2035/36	6.0	199	390
2036/37	0.1	153	731

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-15-01	Replace 4.4 km of trunk main along Weavers Close	2033	4,720
AMP-15-02	Replace 179 lead communication pipes in the Weavers Close area	2032	560
AMP-15-03	New 4.3 Ml storage tank at Station Road	2035	8,010
AMP-15-04	Refurbish Quarrendon No. 1 service reservoir (roof membrane and internal coating)	2031	1,030
AMP-15-05	Rehabilitate 7.4 km of cast-iron distribution mains	2033	3,780
	Total for Quarrendon		**18,100**

For the reservoir defects, the options were refurbishment, replacement on the same site and a new tank on a higher site; refurbishment had the lowest whole-life cost.

Customer contact will be handled by the regional engagement team, with letters sent four weeks before work starts. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.16 Rushmere Down supply zone

The Rushmere Down zone serves approximately 22,200 people in 9,310 properties, centred on an expanding new settlement of Rushmere Down and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 6 service reservoirs with a combined storage of 6.7 megalitres, equivalent to about 2 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	128.8	km	23	1
Distribution mains	364.2	km	8	2
Service reservoirs	6	no.	39	1
Pumping stations	1	no.	74	1
Pressure-reducing valves	37	no.	53	5
Hydrants	1,295	no.	82	5
Customer meters	5,710	no.	33	3
Telemetry outstations	47	no.	44	3

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
RUS-01	3,190	7.4	167	59.7
RUS-02	410	8.5	88	40.2
RUS-03	490	14.7	157	47.8
RUS-04	960	2.5	90	35.6
RUS-05	3,410	14.1	188	35.9
RUS-06	3,410	3.4	156	65.0
RUS-07	1,410	13.7	168	47.7
RUS-08	640	3.1	164	38.8
RUS-09	1,240	7.4	195	51.9

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
RUS-10	3,420	9.4	100	42.6
RUS-11	1,450	3.5	65	62.5

Pressures range from 20 m at the top of Copse Road to 58 m in the lower part of the zone, so the pressure-reducing valves are set to keep night pressures below 45 m.

Condition and performance

Ground movement in the clay soils causes a seasonal peak in bursts in late summer. Over the last three years the zone averaged 10.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 5.97 megalitres per day, and the average customer lost 32.5 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	144	0
Free chlorine residual	207	2
Iron	400	2
Manganese	34	1
Lead	29	0
Coliform bacteria	70	1

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The service reservoirs are close to the end of their structural life and two have known roof defects. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,510 new homes in the zone up to 2036. On current assumptions they will add 15.6 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.8	53	179
2028/29	2.6	142	377
2029/30	2.3	136	174
2030/31	5.0	144	19
2031/32	4.5	201	47
2032/33	4.1	197	362
2033/34	2.0	177	524
2034/35	5.9	180	605

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2035/36	4.3	67	426
2036/37	0.3	129	215

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-16-01	Pressure management in 5 district metered areas	2035	700
AMP-16-02	Upgrade 21 telemetry outstations to cellular communications	2030	300
AMP-16-03	Replace 62 lead communication pipes in the Ropewalk area	2027	160
AMP-16-04	New 3.7 Ml storage tank at Copse Road	2034	6,370
AMP-16-05	Refurbish Rushmere Down No. 1 service reservoir (roof membrane and internal coating)	2035	2,990
	Total for Rushmere Down		**10,520**

Lining rather than replacing the mains was rejected because of the number of service connections, which would each need to be re-made.

Works in this zone are coordinated with the highway authority’s resurfacing programme to avoid digging up new road surfaces. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.17 Stavely Cross supply zone

The Stavely Cross zone serves approximately 35,600 people in 15,750 properties, centred on a coastal resort of Stavely Cross and the surrounding parishes. It is supplied from the Caddon river intake and treatment works and has 4 service reservoirs with a combined storage of 3.8 megalitres, equivalent to about 1 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	51.8	km	83	4
Distribution mains	520.0	km	33	2
Service reservoirs	4	no.	53	4
Pumping stations	1	no.	61	3
Pressure-reducing valves	35	no.	60	3
Hydrants	1,486	no.	28	3
Customer meters	10,927	no.	78	1
Telemetry outstations	10	no.	48	5

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
STA-01	2,160	4.4	112	51.0
STA-02	2,940	2.7	106	67.6
STA-03	1,390	6.1	162	39.7
STA-04	2,410	5.8	61	64.1
STA-05	2,770	4.6	62	56.2

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
STA-06	2,990	4.2	182	45.1
STA-07	1,310	4.4	54	41.2
STA-08	1,690	4.8	146	49.5
STA-09	2,890	1.6	107	63.5

The hydraulic model was recalibrated in 2024 using 44 pressure loggers; it shows that the trunk main along Hollow Way runs close to its capacity on peak days.

Condition and performance

Ground movement in the clay soils causes a seasonal peak in bursts in late summer. Over the last three years the zone averaged 19.1 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 7.91 megalitres per day, and the average customer lost 4.9 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	234	0
Free chlorine residual	148	0
Iron	40	2
Manganese	275	2
Lead	119	0
Coliform bacteria	196	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Telemetry coverage is incomplete, so some reservoir levels are read only during site visits. An agreement with the neighbouring utility allows an emergency cross-connection at Stavely Cross for up to 72 hours. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,620 new homes in the zone up to 2036. On current assumptions they will add 2.2 per cent to peak demand, and the existing network can absorb this growth until about 2029, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	1.7	128	543
2028/29	1.7	27	840
2029/30	1.1	53	218
2030/31	2.5	144	133
2031/32	3.7	198	45
2032/33	0.8	70	39

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2033/34	5.8	186	562
2034/35	2.4	160	153
2035/36	1.2	131	618
2036/37	4.3	113	162

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-17-01	Pressure management in 3 district metered areas	2031	500
AMP-17-02	Refurbish Stavely Cross No. 1 service reservoir (roof membrane and internal coating)	2035	1,890
AMP-17-03	Upgrade 22 telemetry outstations to cellular communications	2029	480
AMP-17-04	Install 19 acoustic leak loggers on strategic mains	2028	50
	Total for Stavely Cross		**2,920**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

The mains renewal will share traffic management with the gas network operator where routes overlap. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.18 Tolland Edge supply zone

The Tolland Edge zone serves approximately 89,100 people in 39,500 properties, centred on a coastal resort of Tolland Edge and the surrounding parishes. It is supplied from the Stavely Cross springs and ultraviolet plant and has 4 service reservoirs with a combined storage of 14.4 megalitres, equivalent to about 1 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	118.4	km	21	4
Distribution mains	146.7	km	79	5
Service reservoirs	4	no.	21	2
Pumping stations	9	no.	13	5
Pressure-reducing valves	16	no.	38	5
Hydrants	2,406	no.	83	5
Customer meters	31,335	no.	53	1
Telemetry outstations	34	no.	62	3

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
TOL-01	1,780	1.5	68	35.5
TOL-02	460	12.8	120	29.4
TOL-03	1,690	8.6	123	66.3
TOL-04	2,320	7.3	82	38.7

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
TOL-05	460	6.0	119	38.8
TOL-06	3,340	11.6	125	60.1
TOL-07	3,160	11.0	63	44.4
TOL-08	700	1.4	74	39.2
TOL-09	2,210	12.0	92	50.3

Night flows have fallen by 26 per cent since the district meters were renewed, but the zone still has three areas with pressures above 70 m.

Condition and performance

The asbestos-cement mains installed in the 1950s and 1960s are now the main source of failures. Over the last three years the zone averaged 9.6 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 2.96 megalitres per day, and the average customer lost 8.5 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	346	0
Free chlorine residual	120	0
Iron	42	3
Manganese	286	3
Lead	46	3
Coliform bacteria	143	0

The zone has a higher than average number of customers on the priority services register, who will be contacted individually before planned interruptions. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

The main risk is the single feed from the source works; an outage of more than eighteen hours would exhaust storage. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 620 new homes in the zone up to 2036. On current assumptions they will add 10.2 per cent to peak demand, and the existing network can absorb this growth until about 2032, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	4.4	85	121
2028/29	1.0	127	733
2029/30	4.1	191	265
2030/31	1.3	30	468
2031/32	1.3	147	789

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2032/33	1.3	157	271
2033/34	5.5	166	760
2034/35	5.9	84	761
2035/36	5.8	73	655
2036/37	5.3	146	831

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-18-01	Rehabilitate 10.1 km of cast-iron distribution mains	2033	6,580
AMP-18-02	Renew booster pumps and controls at Copse Road pumping station	2030	1,450
AMP-18-03	Install 31 acoustic leak loggers on strategic mains	2030	60
AMP-18-04	New 3.4 Ml storage tank at Copse Road	2030	5,530
AMP-18-05	Refurbish Tolland Edge No. 2 service reservoir (roof membrane and internal coating)	2031	1,310
	Total for Tolland Edge		**14,930**

A second feed into the zone was considered, but its cost was not justified by the reduction in risk; additional storage was chosen instead.

The programme assumes that the land purchase for the new pumping station completes in the first year. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.19 Ulverby supply zone

The Ulverby zone serves approximately 65,700 people in 29,320 properties, centred on an expanding new settlement of Ulverby and the surrounding parishes. It is supplied from the Netherfold impounding reservoir and has 4 service reservoirs with a combined storage of 32.0 megalitres, equivalent to about 3 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	118.0	km	42	1
Distribution mains	114.0	km	56	2
Service reservoirs	4	no.	57	4
Pumping stations	9	no.	35	2
Pressure-reducing valves	31	no.	27	4
Hydrants	1,080	no.	40	2
Customer meters	20,871	no.	62	2
Telemetry outstations	35	no.	17	4

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
ULV-01	1,660	2.6	45	35.6
ULV-02	390	12.4	80	39.6

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
ULV-03	500	14.5	41	67.4
ULV-04	1,500	14.6	46	66.2
ULV-05	1,310	8.1	154	62.7
ULV-06	1,660	5.5	165	67.5
ULV-07	2,060	9.0	122	35.5
ULV-08	630	10.9	71	66.7
ULV-09	1,680	12.8	116	55.5
ULV-10	3,380	3.4	96	43.4
ULV-11	3,120	2.4	50	32.6
ULV-12	940	5.9	84	54.3
ULV-13	770	2.7	99	35.9

The hydraulic model was recalibrated in 2023 using 45 pressure loggers; it shows that the trunk main along Beacon Rise runs close to its capacity on peak days.

Condition and performance

The asbestos-cement mains installed in the 1950s and 1960s are now the main source of failures. Over the last three years the zone averaged 12.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 2.95 megalitres per day, and the average customer lost 10.7 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	194	0
Free chlorine residual	138	0
Iron	185	0
Manganese	137	0
Lead	121	2
Coliform bacteria	194	0

Customers in the zone said in the consultation that they would accept longer roadworks if the number of interruptions fell. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Telemetry coverage is incomplete, so some reservoir levels are read only during site visits. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 970 new homes in the zone up to 2036. On current assumptions they will add 7.3 per cent to peak demand, and the existing network can absorb this growth until about 2031, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.1	212	108
2028/29	2.3	16	771
2029/30	5.0	133	698
2030/31	1.3	92	129
2031/32	3.5	120	475
2032/33	5.0	165	120
2033/34	1.7	73	414
2034/35	4.5	157	448
2035/36	0.5	181	98
2036/37	0.8	16	534

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-19-01	Replace 6.1 km of trunk main along Beacon Rise	2028	5,920
AMP-19-02	Renew booster pumps and controls at Copse Road pumping station	2032	1,530
AMP-19-03	New 1.1 Ml storage tank at Copse Road	2028	1,910
AMP-19-04	Refurbish Ulverby No. 3 service reservoir (roof membrane and internal coating)	2033	1,110
	Total for Ulverby		**10,470**

For the reservoir defects, the options were refurbishment, replacement on the same site and a new tank on a higher site; refurbishment had the lowest whole-life cost.

Planning consent is needed for the new reservoir; pre-application discussions have started. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.20 Vexford supply zone

The Vexford zone serves approximately 48,600 people in 20,260 properties, centred on the market town of Vexford and the surrounding parishes. It is supplied from the Netherfold impounding reservoir and has 1 service reservoir with a combined storage of 14.7 megalitres, equivalent to about 2 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	138.4	km	76	2
Distribution mains	499.4	km	52	2
Service reservoirs	1	no.	54	4
Pumping stations	3	no.	51	2
Pressure-reducing valves	48	no.	34	1
Hydrants	660	no.	75	5
Customer meters	13,101	no.	11	4
Telemetry outstations	62	no.	24	3

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
VEX-01	2,450	14.8	114	34.4
VEX-02	2,160	7.5	41	50.0
VEX-03	2,570	13.1	62	53.4
VEX-04	2,000	14.3	46	38.9
VEX-05	1,420	1.5	104	58.3
VEX-06	2,490	8.7	146	43.6
VEX-07	600	11.5	163	52.4
VEX-08	1,080	10.5	80	40.8
VEX-09	860	5.7	163	60.4
VEX-10	1,050	3.5	117	50.3
VEX-11	2,620	7.5	131	43.3

The critical point for pressure is at Tanners Row, where the service level of 20 m is met only when both booster pumps are available.

Condition and performance

Condition is mixed: the town centre has old cast-iron mains, while the outer estates have modern polyethylene pipes. Over the last three years the zone averaged 31.3 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 0.69 megalitres per day, and the average customer lost 31.5 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	223	1
Free chlorine residual	316	0
Iron	203	1
Manganese	327	2
Lead	233	0
Coliform bacteria	393	0

Taste and odour contacts rise every autumn when the source water changes, and are handled by a seasonal treatment adjustment. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Several trunk main crossings under the railway cannot be inspected without possessions. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,180 new homes in the zone up to 2036. On current assumptions they will add 2.2 per cent to peak demand, and the existing network can absorb this growth until about 2029, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.6	15	687
2028/29	4.2	202	36
2029/30	5.3	103	793
2030/31	3.0	39	421
2031/32	5.8	179	768
2032/33	1.0	31	850
2033/34	2.7	166	171
2034/35	3.6	111	58
2035/36	1.7	194	593
2036/37	1.1	103	311

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-20-01	Replace 179 lead communication pipes in the School Lane area	2033	580
AMP-20-02	New 4.4 Ml storage tank at Tanners Row	2033	5,950
AMP-20-03	Replace 1.6 km of trunk main along School Lane	2029	2,050
AMP-20-04	Refurbish Vexford No. 1 service reservoir (roof membrane and internal coating)	2029	1,180
AMP-20-05	Install 44 acoustic leak loggers on strategic mains	2035	170
	Total for Vexford		**9,930**

For the mains, targeted replacement of the worst lengths was compared with blanket renewal of all cast-iron mains; targeted replacement gives most of the benefit at about a third of the cost.

The mains renewal will share traffic management with the gas network operator where routes overlap. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.21 Wrayholme supply zone

The Wrayholme zone serves approximately 91,900 people in 36,900 properties, centred on a commuter town of Wrayholme and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 6 service reservoirs with a combined storage of 17.9 megalitres, equivalent to about 1 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	52.9	km	14	4
Distribution mains	612.9	km	38	2
Service reservoirs	6	no.	68	3
Pumping stations	8	no.	65	3
Pressure-reducing valves	47	no.	45	3
Hydrants	2,489	no.	32	5
Customer meters	30,105	no.	35	2
Telemetry outstations	13	no.	9	2

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
WRA-01	3,280	14.6	76	46.9
WRA-02	320	1.0	68	50.5
WRA-03	1,070	6.0	106	33.4
WRA-04	2,160	8.1	50	62.9
WRA-05	2,790	3.2	127	42.4
WRA-06	2,390	6.1	72	50.0
WRA-07	3,210	5.1	169	64.8
WRA-08	2,990	12.9	165	41.5
WRA-09	3,200	6.4	193	55.7
WRA-10	1,570	13.8	175	35.3
WRA-11	2,140	4.7	123	32.7
WRA-12	2,290	11.8	62	33.3

Pressures range from 23 m at the top of Gallows Hill to 77 m in the lower part of the zone, so the pressure-reducing valves are set to keep night pressures below 45 m.

Condition and performance

Ground movement in the clay soils causes a seasonal peak in bursts in late summer. Over the last three years the zone averaged 27.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 2.30 megalitres per day, and the average customer lost 4.8 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	78	2
Free chlorine residual	151	0
Iron	99	2
Manganese	323	0
Lead	185	0
Coliform bacteria	236	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Several trunk main crossings under the railway cannot be inspected without possessions. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 2,220 new homes in the zone up to 2036. On current assumptions they will add 12.7 per cent to peak demand, and the existing network can absorb this growth until about 2032, after which the

reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	5.6	199	233
2028/29	3.6	9	211
2029/30	4.9	216	33
2030/31	2.2	217	675
2031/32	1.3	7	606
2032/33	3.8	47	112
2033/34	2.6	153	736
2034/35	4.3	129	779
2035/36	2.8	211	739
2036/37	0.3	161	271

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-21-01	Replace 1.3 km of trunk main along Beacon Rise	2028	1,420
AMP-21-02	Refurbish Wrayholme No. 1 service reservoir (roof membrane and internal coating)	2027	1,220
AMP-21-03	Rehabilitate 3.0 km of cast-iron distribution mains	2028	1,430
AMP-21-04	Replace 260 lead communication pipes in the Beacon Rise area	2031	730
AMP-21-05	Install 18 acoustic leak loggers on strategic mains	2035	40
	Total for Wrayholme		**4,840**

For the reservoir defects, the options were refurbishment, replacement on the same site and a new tank on a higher site; refurbishment had the lowest whole-life cost.

The programme assumes that the land purchase for the new pumping station completes in the first year. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.22 Westcombe Rise supply zone

The Westcombe Rise zone serves approximately 30,400 people in 13,410 properties, centred on a former mill town of Westcombe Rise and the surrounding parishes. It is supplied from the Elm Valley boreholes and has 5 service reservoirs with a combined storage of 15.1 megalitres, equivalent to about 4 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	140.4	km	38	5
Distribution mains	67.9	km	52	3
Service reservoirs	5	no.	44	5
Pumping stations	4	no.	21	5
Pressure-reducing valves	21	no.	41	4
Hydrants	1,055	no.	11	1

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Customer meters	11,362	no.	85	5
Telemetry outstations	43	no.	50	2

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
WES-01	3,100	14.5	108	40.4
WES-02	880	14.3	99	59.6
WES-03	3,290	8.6	170	29.3
WES-04	570	1.4	192	54.1
WES-05	3,490	7.7	83	66.8
WES-06	2,480	5.0	51	32.5
WES-07	2,610	6.0	85	48.8
WES-08	3,190	8.0	41	50.1
WES-09	2,620	11.7	167	67.6
WES-10	2,640	10.7	115	60.2

The critical point for pressure is at Sandpit Lane, where the service level of 20 m is met only when both booster pumps are available.

Condition and performance

The asbestos-cement mains installed in the 1950s and 1960s are now the main source of failures. Over the last three years the zone averaged 22.2 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 3.06 megalitres per day, and the average customer lost 8.1 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	220	0
Free chlorine residual	154	0
Iron	88	0
Manganese	265	3
Lead	68	0
Coliform bacteria	397	0

Customer contacts about discoloured water are above the regional average and are concentrated in the older streets. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Several trunk main crossings under the railway cannot be inspected without possessions. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 720 new homes in the zone up to 2036. On current assumptions they will add 11.1 per cent to peak demand, and the existing network can absorb this growth until about 2028, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	1.3	100	789
2028/29	1.2	182	185
2029/30	1.3	26	604
2030/31	4.0	132	454
2031/32	1.5	198	32
2032/33	5.1	133	643
2033/34	3.4	63	334
2034/35	3.1	26	741
2035/36	3.2	109	496
2036/37	1.0	98	596

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-22-01	Upgrade 21 telemetry outstations to cellular communications	2035	410
AMP-22-02	New 1.9 Ml storage tank at Sandpit Lane	2032	3,330
AMP-22-03	Replace 76 lead communication pipes in the Long Furlong area	2033	210
AMP-22-04	Pressure management in 3 district metered areas	2028	580
	Total for Westcombe Rise		**4,530**

The option of doing nothing was costed in each case and used as the baseline for the benefit assessment.

Planning consent is needed for the new reservoir; pre-application discussions have started. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.23 Yarrowfield supply zone

The Yarrowfield zone serves approximately 45,200 people in 19,970 properties, centred on an expanding new settlement of Yarrowfield and the surrounding parishes. It is supplied from a bulk transfer from the Gorsemere trunk main and has 4 service reservoirs with a combined storage of 39.8 megalitres, equivalent to about 6 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	138.7	km	80	5
Distribution mains	592.6	km	42	2
Service reservoirs	4	no.	48	2
Pumping stations	2	no.	55	5
Pressure-reducing valves	16	no.	14	4

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Hydrants	476	no.	21	1
Customer meters	14,481	no.	12	3
Telemetry outstations	30	no.	55	3

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
YAR-01	1,240	12.5	74	65.1
YAR-02	1,480	3.5	53	42.9
YAR-03	440	7.0	168	52.2
YAR-04	1,700	8.3	111	51.8
YAR-05	1,490	12.4	71	59.8
YAR-06	2,180	5.0	196	59.8
YAR-07	2,710	13.7	170	53.5
YAR-08	1,080	3.5	178	54.3
YAR-09	1,910	13.7	54	36.7
YAR-10	2,530	13.3	57	53.3

The hydraulic model was recalibrated in 2025 using 77 pressure loggers; it shows that the trunk main along School Lane runs close to its capacity on peak days.

Condition and performance

Ground movement in the clay soils causes a seasonal peak in bursts in late summer. Over the last three years the zone averaged 28.8 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 9.15 megalitres per day, and the average customer lost 17.7 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	397	0
Free chlorine residual	342	1
Iron	53	0
Manganese	373	2
Lead	279	0
Coliform bacteria	235	1

Complaints about low pressure have fallen since the booster pumps were replaced, but remain high in the upper part of the zone. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Several trunk main crossings under the railway cannot be inspected without possessions. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 470 new homes in the zone up to 2036. On current assumptions they will add 12.4 per cent to peak demand, and the existing network can absorb this growth until about 2030, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	2.5	170	262
2028/29	2.4	34	116
2029/30	5.4	69	713
2030/31	2.2	63	257
2031/32	3.7	11	484
2032/33	1.4	1	895
2033/34	4.2	151	556
2034/35	3.6	105	216
2035/36	5.2	63	261
2036/37	2.4	159	288

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-23-01	Replace 3.7 km of trunk main along School Lane	2028	5,100
AMP-23-02	New 1.6 Ml storage tank at Tanners Row	2034	2,240
AMP-23-03	Renew booster pumps and controls at Tanners Row pumping station	2029	970
AMP-23-04	Refurbish Yarrowfield No. 1 service reservoir (roof membrane and internal coating)	2027	1,310
AMP-23-05	Rehabilitate 11.2 km of cast-iron distribution mains	2031	7,140
	Total for Yarrowfield		**16,760**

For the reservoir defects, the options were refurbishment, replacement on the same site and a new tank on a higher site; refurbishment had the lowest whole-life cost.

The mains renewal will share traffic management with the gas network operator where routes overlap. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

5.24 Yeldham Moss supply zone

The Yeldham Moss zone serves approximately 27,600 people in 12,190 properties, centred on an expanding new settlement of Yeldham Moss and the surrounding parishes. It is supplied from the Caddon river intake and treatment works and has 3 service reservoirs with a combined storage of 23.4 megalitres, equivalent to about 6 hours of average demand.

Asset register summary

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Trunk mains	114.8	km	50	2
Distribution mains	272.1	km	24	1
Service reservoirs	3	no.	84	2

Asset class	Quantity	Unit	Median age (years)	Condition grade (1–5)
Pumping stations	4	no.	79	1
Pressure-reducing valves	44	no.	51	5
Hydrants	1,134	no.	49	5
Customer meters	8,754	no.	83	2
Telemetry outstations	45	no.	12	5

District metered areas

DMA	Properties	Minimum night flow (l/s)	Leakage (l/property/day)	Average pressure (m)
YEL-01	1,620	7.3	96	60.7
YEL-02	2,660	13.7	106	44.1
YEL-03	2,780	9.7	70	52.2
YEL-04	1,070	12.4	148	37.1
YEL-05	1,190	9.9	163	57.9
YEL-06	1,840	6.2	98	29.9
YEL-07	1,600	9.7	196	59.7
YEL-08	2,450	14.1	122	41.4
YEL-09	1,580	8.4	43	41.1
YEL-10	1,990	1.1	74	62.9
YEL-11	1,600	11.2	110	43.9
YEL-12	3,060	9.9	84	61.1

Night flows have fallen by 27 per cent since the district meters were renewed, but the zone still has three areas with pressures above 70 m.

Condition and performance

The zone has a high share of small-diameter galvanised steel mains, which corrode from the inside. Over the last three years the zone averaged 23.9 bursts per 100 km of main per year, against an Authority-wide average of 14.2. Leakage is estimated at 7.22 megalitres per day, and the average customer lost 19.1 minutes of supply per year to unplanned interruptions.

Water quality and customers

Parameter (2025)	Samples	Samples above standard
Turbidity	153	0
Free chlorine residual	250	0
Iron	152	0
Manganese	304	2
Lead	319	3
Coliform bacteria	78	0

Complaints about low pressure have fallen since the booster pumps were replaced, but remain high in the upper part of the zone. Each sample above the standard was investigated, and the cause and the action taken were reported to the drinking water inspectorate.

Risks

Rising peak summer demand from new housing is reducing the margin between demand and pumping capacity. The risk scores for this zone are carried in the corporate risk register and reviewed by the asset strategy group twice a year.

Growth and new connections

The local plan allocates about 1,810 new homes in the zone up to 2036. On current assumptions they will add 2.2 per cent to peak demand, and the existing network can absorb this growth until about 2032, after which the reinforcement listed below is needed. Developers pay for the new mains on their sites and contribute to network reinforcement through infrastructure charges.

Year	Mains renewed (km)	Lead pipes replaced	Meters installed
2027/28	0.5	154	307
2028/29	5.3	23	275
2029/30	0.6	24	610
2030/31	2.1	118	554
2031/32	3.3	68	715
2032/33	1.7	2	514
2033/34	3.0	21	183
2034/35	1.3	96	572
2035/36	4.3	150	685
2036/37	5.5	178	666

Planned interventions 2027–2036

Ref.	Intervention	Start	Cost (£k)
AMP-24-01	Replace 4.5 km of trunk main along Hollow Way	2031	4,210
AMP-24-02	Rehabilitate 4.2 km of cast-iron distribution mains	2030	2,650
AMP-24-03	Install 20 acoustic leak loggers on strategic mains	2030	50
AMP-24-04	Refurbish Yeldham Moss No. 1 service reservoir (roof membrane and internal coating)	2035	2,850
AMP-24-05	Replace 143 lead communication pipes in the Hollow Way area	2029	400
AMP-24-06	Renew booster pumps and controls at Station Road pumping station	2027	1,270
	Total for Yeldham Moss		**11,430**

A second feed into the zone was considered, but its cost was not justified by the reduction in risk; additional storage was chosen instead.

The programme assumes that the land purchase for the new pumping station completes in the first year. Costs are at 2026/27 prices and include design, construction, land and a risk allowance of 15 per cent.

6 Capital programme

The interventions in the zone profiles add up to a ten-year capital programme of about £1,254 million at 2026/27 prices. The table shows the programme by year and by category of asset. Treatment works schemes are planned separately but are shown here so that the whole programme is in one place.

Year	Mains (£k)	Storage (£k)	Pumping (£k)	Treatment (£k)	Telemetry (£k)	Total (£k)
2027/28	54,770	27,010	21,530	10,370	3,810	117,490
2028/29	66,780	47,200	23,110	6,500	4,270	147,860
2029/30	69,280	40,010	17,930	12,200	5,950	145,370
2030/31	43,610	29,130	10,350	9,750	6,550	99,390
2031/32	71,300	27,380	14,190	8,250	3,950	125,070
2032/33	48,240	31,230	9,150	13,340	6,650	108,610
2033/34	56,280	40,550	14,470	8,330	7,190	126,820
2034/35	50,270	43,030	16,990	11,630	7,120	129,040
2035/36	53,990	28,790	14,500	11,050	3,050	111,380
2036/37	69,920	29,030	25,320	10,400	8,710	143,380

The programme is front-loaded towards storage in the first five years because of the undertakings agreed with the drinking water inspectorate. Mains renewal rises steadily to reach its full rate by 2030, which gives the contractor framework time to build up capacity.

Schemes with a value above £5 million have been through an independent cost review. The review found that the estimates were reasonable but that the risk allowance for two reservoir schemes was too low; those allowances have been raised from 15 to 25 per cent.

About 12 per cent of the programme is not yet assigned to named schemes. It is held as a reactive allowance for failures that cannot be predicted, such as a trunk main burst on a crossing, and is released by the investment committee when a need arises.

7 Operating costs and financing

Operating costs are affected by the plan in two ways. New assets such as storage tanks and pressure management schemes add maintenance and energy costs, while replacing old pumps and reducing leakage lowers them. The net effect is estimated as a saving of £1.9 million a year by 2036.

The table below compares the operating budget for 2025/26 with the outturn, by cost category. Positive variances are underspends; negative variances are overspends. The largest variance is in energy, where the price agreed for 2025/26 was higher than assumed when the budget was set.

Cost category	Budget (£k)	Outturn (£k)	Variance (£k)
Employee costs	18,420	18,105	315
Energy	4,820	6,070	1,250
Chemicals	2,140	2,385	−245
Materials and spares	3,310	3,195	115
Contractors and hired services	7,650	7,920	−270
Business rates	2,905	2,905	0
Insurance	1,210	1,340	−130
IT and telemetry licences	1,480	1,395	85
Vehicles and plant	1,960	2,010	−50
Abstraction and discharge charges	980	1,015	−35

Energy is the most volatile operating cost. The Authority buys about 60 per cent of its electricity under a fixed-price contract and the rest on the wholesale market. The plan assumes that the fixed share rises to 80 per cent from 2028, and that the solar arrays at three treatment works supply 6 per cent of demand.

The capital programme will be financed from a mix of retained revenue, borrowing and developer contributions. Gearing is forecast to stay below 60 per cent throughout the planning period. The financial plan assumes the cost of debt set in the regulator’s final determination.

The Authority will continue to seek external funding for schemes with wider benefits, such as catchment management and flood protection that also protects other infrastructure. No external funding is assumed in the figures in this chapter.

8 Risk and assurance

The main risks to delivering the plan are recorded in the risk register below. Each risk is scored for likelihood and impact on a scale of 1 to 5. The register is reviewed every quarter by the executive team and twice a year by the audit and risk committee.

Ref.	Risk	Likelihood	Impact	Main controls
R-01	Prolonged outage at the Brackwater treatment works	4	5	Second treatment stream; mobile treatment unit contract
R-02	Drought reduces yield of the Elm Valley boreholes	3	4	Drought plan triggers; demand restrictions; bulk transfer
R-03	Failure of a critical trunk main crossing the railway	2	5	Inspection under possession; valve exercising; spare pipe stock
R-04	Cyber attack on the telemetry and control system	3	5	Network segregation; incident exercises; offline backups
R-05	Contractor capacity shortfall in peak years	4	3	Framework with three contractors; programme smoothing
R-06	Construction cost inflation above forecast	4	3	Risk allowance; quarterly re-forecasting; value engineering
R-07	Flooding of low-lying pumping stations	3	4	Flood walls at four sites; kiosk relocation; portable pumps
R-08	Loss of power at strategic sites	3	4	Standby generators; fuel contracts; load shedding plan
R-09	Water quality failure from reservoir ingress	2	5	Roof inspections; hatch security; sampling regime
R-10	Unplanned growth from housing allocations	3	3	Liaison with planning authorities; staged reinforcement
R-11	Loss of key operational staff	3	3	Succession plans; apprenticeships; knowledge capture
R-12	Delay in planning consent for new storage	3	3	Early engagement; alternative sites identified
R-13	Third-party damage to mains during street works	4	2	Records sharing; site supervision; recovery of costs
R-14	Failure of legacy billing interface for meter data	2	3	Interface replacement in 2028; manual fallback

Ref.	Risk	Likelihood	Impact	Main controls
R-15	Pollution incident at an abstraction point	2	4	Catchment monitoring; intake shut-off; alternative sources

The plan has been assured in three ways. The cost estimates were reviewed by an independent cost consultant; the asset data and condition assessments were audited by the internal audit team; and the prioritisation method was reviewed by an external panel of asset management practitioners. Their findings and the Authority's responses are available on request.

Data quality remains the main limitation of the plan. The asset register still contains about 4,200 mains with an unknown installation date, and the condition of buried assets is mostly inferred from bursts rather than inspected. The Authority will continue its sampling programme of pipe condition assessments to improve the data.

9 Delivery and monitoring

Delivery of the plan will be monitored against the key performance indicators below, which are aligned with the performance commitments agreed with the regulator. Progress will be reported to the Board every six months and published in the annual performance report.

Indicator	2025/26 actual	2031 target	2036 target
Leakage (Ml/d, three-year average)	38.4	33.0	28.5
Bursts per 1,000 km of main	142	130	118
Average minutes lost per property per year	11.6	8.0	5.0
Properties with low pressure (no.)	412	250	120
Lead communication pipes replaced (cumulative)	1,860	6,500	14,000
Service reservoirs at condition grade 4 or 5 (no.)	17	9	3
Customer contacts about water quality per 1,000 population	1.42	1.20	1.00
Telemetry outstations on cellular communications (%)	36	75	100

If an indicator falls behind its trajectory for two consecutive reports, the Asset Strategy Group will prepare a recovery plan for the executive team. The plan will be reviewed in full every five years, and earlier if there is a significant change in regulation, demand or funding.

Appendix A Glossary

Asset class
A group of assets of the same kind that are planned and reported together, such as trunk mains or service reservoirs.

Burst
A failure of a water main that causes a visible leak or loss of supply and needs a repair.

Capital maintenance
Expenditure that restores or replaces existing assets so that they keep delivering the same service.

Communication pipe
The section of service pipe between the water main and the boundary of the customer's property.

Condition grade
A score from 1 (very good) to 5 (very poor) that describes the physical state of an asset.

Critical asset
An asset whose failure would interrupt supply to more than 5,000 properties or cause a serious safety risk.

District metered area (DMA)
A section of network with metered inflows and outflows, used to measure leakage.

Enhancement
Expenditure that improves service or adds capacity beyond what the existing assets provide.

Headroom
The margin between available supply and forecast demand, including an allowance for uncertainty.

Interruption
A period during which a property receives no water, or water at a pressure too low to use.

Lead communication pipe
A communication pipe made of lead, which can raise lead levels at the customer’s tap.

Megalitre (Ml)
One million litres.

Non-revenue water
Water put into supply that is not billed, mainly leakage and unbilled authorised use.

Outstation
A telemetry unit on site that reports levels, flows, alarms and pump states to the control room.

Pressure management
Reducing pressure at times of low demand to cut leakage and the number of bursts.

Resilience
The ability to keep supplying customers during and after shocks such as floods, power cuts or droughts.

Risk allowance
An amount added to cost estimates to cover risks that are known but cannot yet be priced.

Service reservoir
A covered tank that stores treated water close to customers and balances daily demand.

Trunk main
A large-diameter main that carries treated water from works to service reservoirs or between zones.

Whole-life cost
The total cost of owning an asset over its life, including construction, operation, maintenance and disposal.

Appendix B Mains renewal schedule 2027–2031

The schedule lists the mains renewal schemes planned for the first five years. Lengths are design estimates and may change by up to 10 per cent at detailed design. Material codes: CI cast iron, AC asbestos cement, GS galvanised steel, DI ductile iron, PVC unplasticised PVC.

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0001	Ardley Fen	Chapel Hill to Weavers Close	GS	225	1,350	2027	Slip-lining
MR-0002	Ardley Fen	Long Furlong	DI	300	1,900	2027	Pipe bursting
MR-0003	Ardley Fen	Ferry Road crossing	AC	450	330	2027	Directional drilling

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0004	Ardley Fen	Mill Lane (south)	AC	180	1,250	2027	Open cut
MR-0005	Brackwater	Orchard Way (south)	CI	100	500	2027	Directional drilling
MR-0006	Brackwater	Sandpit Lane (south)	DI	250	2,260	2027	Open cut
MR-0007	Brackwater	Weavers Close crossing	GS	225	2,120	2027	Directional drilling
MR-0008	Brackwater	Bridge End (north)	CI	100	940	2027	Open cut
MR-0009	Brackwater	Bridge End	DI	250	110	2027	Directional drilling
MR-0010	Caddon Heath	Market Place to Quarry Lane	AC	355	2,350	2027	Slip-lining
MR-0011	Caddon Heath	Orchard Way (south)	PVC	125	1,460	2027	Slip-lining
MR-0012	Caddon Heath	Kiln Street (south)	CI	450	900	2027	Slip-lining
MR-0013	Caddon Heath	Orchard Way to Copse Road	CI	250	510	2027	Directional drilling
MR-0014	Caddon Heath	Quarry Lane to Copse Road	CI	300	2,100	2027	Slip-lining
MR-0015	Dunholt	Orchard Way (west)	GS	250	1,150	2027	Open cut
MR-0016	Dunholt	Ferry Road (south)	PVC	450	1,460	2027	Open cut
MR-0017	Dunholt	Hollow Way (east)	AC	450	2,120	2027	Open cut
MR-0018	Dunholt	Station Road to Orchard Way	AC	150	550	2027	Slip-lining
MR-0019	Dunholt	Weavers Close (east)	DI	300	2,300	2027	Pipe bursting
MR-0020	Elmscar	Beacon Rise	CI	450	350	2027	Directional drilling
MR-0021	Elmscar	Station Road to Tanners Row	DI	75	210	2027	Open cut
MR-0022	Elmscar	Gallows Hill (north)	AC	100	1,580	2027	Open cut
MR-0023	Fernlow	Gallows Hill crossing	CI	150	1,820	2027	Open cut
MR-0024	Fernlow	Tanners Row to Hollow Way	PVC	355	330	2027	Directional drilling
MR-0025	Fernlow	Ropewalk (north)	GS	150	950	2027	Open cut
MR-0026	Fernlow	Mill Lane (west)	AC	300	330	2027	Directional drilling
MR-0027	Fernlow	Copse Road (west)	DI	150	1,410	2027	Open cut
MR-0028	Fernlow	Beacon Rise crossing	AC	100	750	2027	Slip-lining
MR-0029	Fernlow	Sandpit Lane (west)	DI	450	1,760	2027	Directional drilling
MR-0030	Gorsemere	Tanners Row (south)	AC	100	340	2027	Open cut
MR-0031	Gorsemere	Mill Lane (west)	PVC	225	1,470	2027	Pipe bursting

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0032	Gorsemere	Kiln Street crossing	DI	75	460	2027	Open cut
MR-0033	Gorsemere	Quarry Lane (west)	DI	90	1,870	2027	Open cut
MR-0034	Gorsemere	Copse Road (north)	CI	250	2,190	2027	Open cut
MR-0035	Gorsemere	Quarry Lane	CI	75	1,070	2027	Open cut
MR-0036	Gorsemere	Ferry Road (south)	PVC	300	1,180	2027	Directional drilling
MR-0037	Gorsemere	Copse Road (south)	PVC	75	1,150	2027	Slip-lining
MR-0038	Gorsemere	Hollow Way (north)	GS	225	1,170	2027	Open cut
MR-0039	Hatherby	Bridge End to Tanners Row	CI	180	1,530	2027	Directional drilling
MR-0040	Hatherby	Orchard Way (north)	PVC	450	2,160	2027	Open cut
MR-0041	Hatherby	Quarry Lane	CI	100	560	2027	Open cut
MR-0042	Hatherby	Ropewalk crossing	PVC	250	500	2027	Pipe bursting
MR-0043	Hatherby	Sandpit Lane crossing	AC	180	2,190	2027	Directional drilling
MR-0044	Hatherby	Market Place (south)	CI	250	1,190	2027	Directional drilling
MR-0045	Ivelstone	Long Furlong (east)	CI	150	1,640	2027	Open cut
MR-0046	Ivelstone	Bridge End to Hollow Way	GS	90	340	2027	Open cut
MR-0047	Ivelstone	Tanners Row (south)	AC	450	220	2027	Slip-lining
MR-0048	Ivelstone	Orchard Way (north)	GS	125	680	2027	Open cut
MR-0049	Ivelstone	Quarry Lane (north)	CI	150	1,130	2027	Slip-lining
MR-0050	Kelmarsh Vale	Orchard Way (north)	CI	250	930	2027	Pipe bursting
MR-0051	Kelmarsh Vale	High Causeway (east)	CI	180	540	2027	Open cut
MR-0052	Kelmarsh Vale	Bridge End (west)	GS	300	140	2027	Pipe bursting
MR-0053	Kelmarsh Vale	School Lane (west)	CI	355	2,080	2027	Slip-lining
MR-0054	Kelmarsh Vale	Station Road	AC	300	1,010	2027	Slip-lining
MR-0055	Kelmarsh Vale	Copse Road to Bridge End	GS	75	2,260	2027	Open cut
MR-0056	Kelmarsh Vale	Gallows Hill (south)	CI	100	1,140	2027	Open cut
MR-0057	Kelmarsh Vale	Sandpit Lane (north)	CI	75	1,720	2027	Pipe bursting
MR-0058	Larkhollow	Kiln Street	CI	180	230	2027	Directional drilling
MR-0059	Larkhollow	Quarry Lane (west)	DI	225	1,390	2027	Directional drilling
MR-0060	Larkhollow	Mill Lane (north)	CI	450	1,710	2027	Open cut
MR-0061	Larkhollow	Weavers Close (south)	AC	450	1,570	2027	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0062	Marrowby	Tanners Row (west)	CI	100	840	2027	Directional drilling
MR-0063	Marrowby	Long Furlong to Quarry Lane	GS	150	1,610	2027	Open cut
MR-0064	Marrowby	Quarry Lane crossing	GS	100	1,870	2027	Slip-lining
MR-0065	Netherfold	Sandpit Lane to Mill Lane	AC	90	1,210	2027	Open cut
MR-0066	Netherfold	Kiln Street (north)	CI	150	300	2027	Slip-lining
MR-0067	Netherfold	Station Road (north)	AC	450	1,420	2027	Slip-lining
MR-0068	Netherfold	Tanners Row crossing	CI	225	1,150	2027	Open cut
MR-0069	Netherfold	Gallows Hill crossing	CI	250	1,180	2027	Open cut
MR-0070	Netherfold	Market Place (east)	PVC	225	120	2027	Open cut
MR-0071	Netherfold	Long Furlong (south)	CI	90	1,610	2027	Open cut
MR-0072	Netherfold	Copse Road (east)	AC	90	1,960	2027	Slip-lining
MR-0073	Pellow Bay	Beacon Rise crossing	CI	125	1,800	2027	Open cut
MR-0074	Pellow Bay	Gallows Hill (south)	DI	90	290	2027	Pipe bursting
MR-0075	Pellow Bay	Market Place (south)	AC	450	1,320	2027	Pipe bursting
MR-0076	Pellow Bay	School Lane to Ferry Road	PVC	180	950	2027	Directional drilling
MR-0077	Quarrendon	Quarry Lane crossing	CI	90	590	2027	Directional drilling
MR-0078	Quarrendon	Weavers Close crossing	AC	300	1,840	2027	Slip-lining
MR-0079	Quarrendon	Station Road	AC	180	1,200	2027	Open cut
MR-0080	Quarrendon	Hollow Way crossing	PVC	100	1,140	2027	Open cut
MR-0081	Quarrendon	Copse Road (south)	AC	225	450	2027	Slip-lining
MR-0082	Quarrendon	Long Furlong (west)	CI	450	1,420	2027	Directional drilling
MR-0083	Quarrendon	Ropewalk to School Lane	CI	250	150	2027	Open cut
MR-0084	Rushmere Down	Kiln Street (north)	AC	100	1,580	2027	Pipe bursting
MR-0085	Rushmere Down	Sandpit Lane (north)	PVC	225	1,750	2027	Slip-lining
MR-0086	Rushmere Down	Tanners Row (south)	CI	225	1,910	2027	Open cut
MR-0087	Rushmere Down	Market Place crossing	PVC	125	570	2027	Open cut
MR-0088	Rushmere Down	Sandpit Lane (south)	AC	300	750	2027	Pipe bursting
MR-0089	Rushmere Down	Weavers Close (north)	GS	150	1,190	2027	Directional drilling

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0090	Stavely Cross	School Lane (east)	PVC	250	1,110	2027	Open cut
MR-0091	Stavely Cross	Beacon Rise (east)	AC	100	2,250	2027	Pipe bursting
MR-0092	Stavely Cross	Copse Road (west)	CI	355	2,240	2027	Open cut
MR-0093	Tolland Edge	Orchard Way	AC	125	1,900	2027	Open cut
MR-0094	Tolland Edge	Tanners Row to Bridge End	CI	100	1,720	2027	Slip-lining
MR-0095	Tolland Edge	Tanners Row (east)	PVC	250	670	2027	Open cut
MR-0096	Tolland Edge	Tanners Row crossing	GS	300	2,350	2027	Directional drilling
MR-0097	Tolland Edge	High Causeway	CI	250	1,940	2027	Slip-lining
MR-0098	Tolland Edge	School Lane (west)	DI	75	1,890	2027	Slip-lining
MR-0099	Ulverby	Chapel Hill (west)	DI	90	2,230	2027	Directional drilling
MR-0100	Ulverby	Ferry Road crossing	AC	150	410	2027	Slip-lining
MR-0101	Ulverby	Quarry Lane crossing	PVC	300	150	2027	Slip-lining
MR-0102	Vexford	Chapel Hill (east)	PVC	90	2,100	2027	Pipe bursting
MR-0103	Vexford	Beacon Rise (north)	CI	75	360	2027	Slip-lining
MR-0104	Wrayholme	Kiln Street (west)	GS	125	2,010	2027	Open cut
MR-0105	Wrayholme	Hollow Way (south)	AC	355	1,390	2027	Directional drilling
MR-0106	Westcombe Rise	Beacon Rise (north)	CI	125	490	2027	Pipe bursting
MR-0107	Westcombe Rise	Copse Road (south)	CI	125	1,180	2027	Directional drilling
MR-0108	Yarrowfield	Station Road (south)	PVC	225	1,220	2027	Directional drilling
MR-0109	Yarrowfield	Tanners Row (west)	PVC	75	1,460	2027	Directional drilling
MR-0110	Yarrowfield	Bridge End crossing	AC	450	1,940	2027	Open cut
MR-0111	Yarrowfield	Sandpit Lane	CI	300	110	2027	Open cut
MR-0112	Yarrowfield	Beacon Rise (north)	AC	225	1,600	2027	Slip-lining
MR-0113	Yeldham Moss	Hollow Way crossing	CI	450	1,050	2027	Open cut
MR-0114	Yeldham Moss	Station Road crossing	CI	75	430	2027	Open cut
MR-0115	Yeldham Moss	School Lane (south)	PVC	125	1,200	2027	Open cut
MR-0116	Yeldham Moss	Beacon Rise (west)	CI	75	1,050	2027	Pipe bursting
MR-0117	Yeldham Moss	Copse Road (north)	CI	90	1,240	2027	Pipe bursting
MR-0118	Ardley Fen	Quarry Lane (east)	DI	300	1,030	2028	Open cut
MR-0119	Ardley Fen	Gallows Hill (west)	AC	355	630	2028	Open cut
MR-0120	Ardley Fen	Copse Road (west)	CI	450	1,190	2028	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0121	Ardley Fen	Station Road to Tanners Row	AC	300	2,010	2028	Slip-lining
MR-0122	Ardley Fen	Beacon Rise	CI	125	990	2028	Slip-lining
MR-0123	Brackwater	Weavers Close (east)	DI	225	240	2028	Open cut
MR-0124	Brackwater	Gallows Hill (east)	CI	100	2,150	2028	Open cut
MR-0125	Brackwater	Gallows Hill to Bridge End	CI	300	530	2028	Open cut
MR-0126	Brackwater	Gallows Hill crossing	CI	450	1,610	2028	Open cut
MR-0127	Brackwater	Ferry Road crossing	CI	125	1,220	2028	Open cut
MR-0128	Brackwater	Tanners Row (north)	DI	90	490	2028	Slip-lining
MR-0129	Caddon Heath	Weavers Close (east)	DI	450	1,440	2028	Pipe bursting
MR-0130	Caddon Heath	Chapel Hill (east)	PVC	250	590	2028	Slip-lining
MR-0131	Caddon Heath	Copse Road	AC	90	1,900	2028	Open cut
MR-0132	Caddon Heath	Weavers Close to School Lane	GS	150	1,460	2028	Open cut
MR-0133	Dunholt	Ferry Road (west)	PVC	100	1,340	2028	Directional drilling
MR-0134	Dunholt	Station Road to Sandpit Lane	GS	225	1,650	2028	Slip-lining
MR-0135	Dunholt	Market Place	CI	355	560	2028	Slip-lining
MR-0136	Elmscar	Hollow Way	CI	100	340	2028	Open cut
MR-0137	Elmscar	Quarry Lane (east)	AC	180	2,180	2028	Open cut
MR-0138	Elmscar	School Lane (west)	CI	150	2,360	2028	Directional drilling
MR-0139	Elmscar	Bridge End (east)	PVC	300	490	2028	Open cut
MR-0140	Fernlow	Market Place to Beacon Rise	GS	300	1,220	2028	Directional drilling
MR-0141	Fernlow	Quarry Lane crossing	GS	225	2,260	2028	Slip-lining
MR-0142	Fernlow	Kiln Street to Weavers Close	PVC	450	560	2028	Pipe bursting
MR-0143	Gorsemere	Ferry Road to Kiln Street	PVC	150	2,070	2028	Open cut
MR-0144	Gorsemere	Sandpit Lane	AC	150	1,240	2028	Open cut
MR-0145	Hatherby	Ferry Road	AC	225	230	2028	Directional drilling
MR-0146	Hatherby	Long Furlong (north)	AC	150	810	2028	Open cut
MR-0147	Hatherby	Mill Lane (west)	CI	90	700	2028	Open cut
MR-0148	Hatherby	Ferry Road	AC	355	1,710	2028	Pipe bursting
MR-0149	Hatherby	Copse Road (west)	CI	75	1,150	2028	Open cut
MR-0150	Hatherby	Orchard Way	PVC	450	370	2028	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0151	Hatherby	Mill Lane to Ferry Road	CI	75	310	2028	Slip-lining
MR-0152	Hatherby	Long Furlong crossing	GS	250	860	2028	Open cut
MR-0153	Ivelstone	Chapel Hill (east)	CI	180	1,750	2028	Open cut
MR-0154	Ivelstone	Ropewalk (north)	GS	225	1,920	2028	Open cut
MR-0155	Ivelstone	Ferry Road to Gallows Hill	DI	225	1,860	2028	Open cut
MR-0156	Ivelstone	Ferry Road crossing	CI	250	690	2028	Pipe bursting
MR-0157	Ivelstone	Beacon Rise (east)	CI	450	1,540	2028	Open cut
MR-0158	Ivelstone	Market Place (north)	CI	225	2,250	2028	Slip-lining
MR-0159	Ivelstone	Beacon Rise (north)	CI	355	530	2028	Open cut
MR-0160	Kelmarsh Vale	Tanners Row (east)	CI	450	1,700	2028	Slip-lining
MR-0161	Kelmarsh Vale	Kiln Street to High Causeway	DI	355	570	2028	Open cut
MR-0162	Kelmarsh Vale	Station Road (north)	PVC	100	910	2028	Open cut
MR-0163	Kelmarsh Vale	Beacon Rise (west)	PVC	300	980	2028	Open cut
MR-0164	Kelmarsh Vale	Tanners Row to School Lane	CI	100	300	2028	Open cut
MR-0165	Kelmarsh Vale	Market Place to High Causeway	PVC	180	1,760	2028	Slip-lining
MR-0166	Larkhollow	Tanners Row (west)	PVC	300	140	2028	Pipe bursting
MR-0167	Larkhollow	Tanners Row to Chapel Hill	GS	90	1,700	2028	Slip-lining
MR-0168	Larkhollow	Bridge End (west)	CI	355	800	2028	Slip-lining
MR-0169	Larkhollow	Chapel Hill (south)	AC	450	1,020	2028	Pipe bursting
MR-0170	Larkhollow	Quarry Lane (south)	CI	250	1,700	2028	Directional drilling
MR-0171	Larkhollow	Mill Lane (east)	PVC	125	200	2028	Open cut
MR-0172	Larkhollow	Kiln Street (west)	CI	125	1,460	2028	Directional drilling
MR-0173	Larkhollow	Market Place (west)	AC	355	2,110	2028	Pipe bursting
MR-0174	Marrowby	Gallows Hill	CI	450	1,080	2028	Slip-lining
MR-0175	Marrowby	Chapel Hill (south)	AC	150	1,780	2028	Open cut
MR-0176	Marrowby	Station Road (west)	AC	250	1,450	2028	Open cut
MR-0177	Marrowby	Mill Lane (south)	GS	450	1,090	2028	Open cut
MR-0178	Marrowby	Tanners Row (north)	AC	225	1,180	2028	Open cut
MR-0179	Netherfold	Ropewalk crossing	AC	150	1,000	2028	Directional drilling
MR-0180	Netherfold	Quarry Lane (south)	CI	75	1,320	2028	Pipe bursting

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0181	Netherfold	High Causeway to Kiln Street	CI	125	300	2028	Open cut
MR-0182	Pellow Bay	High Causeway crossing	AC	250	720	2028	Pipe bursting
MR-0183	Pellow Bay	Market Place to Ferry Road	PVC	180	240	2028	Open cut
MR-0184	Pellow Bay	Sandpit Lane to Kiln Street	AC	300	960	2028	Open cut
MR-0185	Pellow Bay	Ferry Road (east)	GS	450	1,400	2028	Open cut
MR-0186	Pellow Bay	Long Furlong to Mill Lane	AC	225	1,020	2028	Slip-lining
MR-0187	Pellow Bay	High Causeway	PVC	225	2,240	2028	Open cut
MR-0188	Quarrendon	Tanners Row (south)	AC	100	690	2028	Directional drilling
MR-0189	Quarrendon	Quarry Lane (south)	CI	100	1,810	2028	Slip-lining
MR-0190	Quarrendon	Market Place crossing	CI	125	830	2028	Open cut
MR-0191	Quarrendon	Ferry Road (east)	CI	100	840	2028	Slip-lining
MR-0192	Quarrendon	Copse Road (east)	GS	150	1,790	2028	Directional drilling
MR-0193	Rushmere Down	Gallows Hill (south)	DI	225	890	2028	Pipe bursting
MR-0194	Rushmere Down	Hollow Way (west)	CI	225	1,470	2028	Open cut
MR-0195	Rushmere Down	Ropewalk	CI	355	890	2028	Pipe bursting
MR-0196	Rushmere Down	Quarry Lane (north)	CI	125	2,040	2028	Pipe bursting
MR-0197	Rushmere Down	Ropewalk to Long Furlong	CI	150	2,350	2028	Open cut
MR-0198	Rushmere Down	Sandpit Lane (west)	CI	180	1,280	2028	Slip-lining
MR-0199	Rushmere Down	Kiln Street (south)	CI	300	1,250	2028	Open cut
MR-0200	Rushmere Down	Station Road (west)	PVC	225	1,880	2028	Slip-lining
MR-0201	Rushmere Down	Sandpit Lane (south)	DI	150	2,250	2028	Directional drilling
MR-0202	Rushmere Down	Long Furlong (west)	AC	75	1,840	2028	Open cut
MR-0203	Stavely Cross	Beacon Rise (west)	AC	300	120	2028	Directional drilling
MR-0204	Stavely Cross	Chapel Hill (north)	AC	90	210	2028	Slip-lining
MR-0205	Stavely Cross	Orchard Way (north)	PVC	90	590	2028	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0206	Stavely Cross	Chapel Hill (north)	PVC	250	1,910	2028	Open cut
MR-0207	Stavely Cross	Beacon Rise (south)	CI	180	1,010	2028	Open cut
MR-0208	Stavely Cross	Tanners Row (north)	CI	355	120	2028	Open cut
MR-0209	Stavely Cross	Market Place (north)	CI	90	2,080	2028	Open cut
MR-0210	Ulverby	Quarry Lane	AC	90	1,060	2028	Open cut
MR-0211	Ulverby	Bridge End (west)	GS	450	1,890	2028	Directional drilling
MR-0212	Ulverby	Chapel Hill crossing	GS	355	1,010	2028	Directional drilling
MR-0213	Ulverby	Hollow Way (west)	CI	75	470	2028	Directional drilling
MR-0214	Vexford	Ferry Road to Market Place	GS	225	1,190	2028	Slip-lining
MR-0215	Vexford	Copse Road to Bridge End	AC	355	380	2028	Directional drilling
MR-0216	Vexford	Orchard Way (south)	GS	150	130	2028	Directional drilling
MR-0217	Vexford	Tanners Row crossing	AC	125	870	2028	Open cut
MR-0218	Vexford	Tanners Row crossing	CI	225	1,170	2028	Slip-lining
MR-0219	Wrayholme	Tanners Row	CI	75	1,430	2028	Open cut
MR-0220	Wrayholme	Long Furlong (west)	PVC	225	1,150	2028	Slip-lining
MR-0221	Wrayholme	Kiln Street (north)	CI	225	840	2028	Pipe bursting
MR-0222	Wrayholme	Kiln Street (north)	GS	100	1,060	2028	Pipe bursting
MR-0223	Wrayholme	Station Road (west)	DI	125	1,730	2028	Pipe bursting
MR-0224	Westcombe Rise	Mill Lane (north)	PVC	150	350	2028	Open cut
MR-0225	Westcombe Rise	Long Furlong (north)	DI	150	1,280	2028	Directional drilling
MR-0226	Westcombe Rise	Station Road (east)	AC	250	700	2028	Open cut
MR-0227	Westcombe Rise	Market Place (north)	DI	180	1,620	2028	Pipe bursting
MR-0228	Westcombe Rise	High Causeway (south)	PVC	250	1,040	2028	Open cut
MR-0229	Westcombe Rise	Hollow Way crossing	CI	355	420	2028	Open cut
MR-0230	Yarrowfield	Bridge End (south)	AC	225	1,300	2028	Directional drilling
MR-0231	Yarrowfield	Sandpit Lane crossing	GS	180	1,950	2028	Directional drilling
MR-0232	Yarrowfield	Sandpit Lane (west)	AC	450	1,440	2028	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0233	Yarrowfield	School Lane (west)	CI	125	2,280	2028	Directional drilling
MR-0234	Yarrowfield	Quarry Lane crossing	DI	250	150	2028	Open cut
MR-0235	Yeldham Moss	Weavers Close (south)	CI	180	1,320	2028	Slip-lining
MR-0236	Yeldham Moss	Ropewalk to High Causeway	PVC	100	1,980	2028	Open cut
MR-0237	Yeldham Moss	Hollow Way (north)	CI	125	1,110	2028	Slip-lining
MR-0238	Yeldham Moss	School Lane crossing	DI	450	440	2028	Slip-lining
MR-0239	Yeldham Moss	Market Place crossing	CI	300	1,100	2028	Open cut
MR-0240	Ardley Fen	Station Road (east)	GS	150	1,640	2029	Slip-lining
MR-0241	Ardley Fen	Bridge End (south)	AC	225	1,860	2029	Open cut
MR-0242	Brackwater	Beacon Rise	PVC	150	390	2029	Slip-lining
MR-0243	Brackwater	Orchard Way to Copse Road	CI	75	860	2029	Open cut
MR-0244	Brackwater	Beacon Rise to Gallows Hill	AC	300	1,870	2029	Open cut
MR-0245	Caddon Heath	Orchard Way (south)	DI	450	930	2029	Slip-lining
MR-0246	Caddon Heath	Station Road crossing	CI	150	700	2029	Slip-lining
MR-0247	Caddon Heath	Long Furlong (east)	AC	90	1,420	2029	Open cut
MR-0248	Caddon Heath	Sandpit Lane (south)	CI	125	1,690	2029	Pipe bursting
MR-0249	Caddon Heath	Tanners Row (east)	CI	300	1,330	2029	Open cut
MR-0250	Dunholt	Chapel Hill	CI	300	410	2029	Open cut
MR-0251	Dunholt	Long Furlong (east)	CI	100	1,640	2029	Open cut
MR-0252	Dunholt	Quarry Lane (north)	GS	90	1,730	2029	Slip-lining
MR-0253	Dunholt	Kiln Street (north)	AC	90	1,570	2029	Slip-lining
MR-0254	Elmscar	Orchard Way crossing	CI	100	190	2029	Open cut
MR-0255	Elmscar	Weavers Close (west)	CI	180	400	2029	Slip-lining
MR-0256	Elmscar	Bridge End (north)	PVC	150	1,950	2029	Open cut
MR-0257	Elmscar	Bridge End (west)	GS	250	2,250	2029	Open cut
MR-0258	Elmscar	Market Place to Ferry Road	GS	225	1,600	2029	Open cut
MR-0259	Fernlow	Quarry Lane (west)	CI	150	920	2029	Slip-lining
MR-0260	Fernlow	Station Road crossing	GS	125	1,630	2029	Slip-lining
MR-0261	Fernlow	School Lane	DI	100	1,390	2029	Slip-lining
MR-0262	Fernlow	Sandpit Lane (north)	AC	75	1,760	2029	Open cut
MR-0263	Fernlow	Beacon Rise (north)	CI	180	1,130	2029	Directional drilling
MR-0264	Fernlow	Mill Lane (north)	DI	450	190	2029	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0265	Fernlow	Station Road (north)	CI	75	1,090	2029	Open cut
MR-0266	Fernlow	Tanners Row (south)	CI	100	2,240	2029	Pipe bursting
MR-0267	Gorsemere	Bridge End (east)	PVC	300	2,180	2029	Open cut
MR-0268	Gorsemere	Orchard Way to Long Furlong	AC	355	700	2029	Open cut
MR-0269	Hatherby	Kiln Street (north)	PVC	150	1,060	2029	Directional drilling
MR-0270	Hatherby	High Causeway (west)	AC	100	510	2029	Slip-lining
MR-0271	Hatherby	Mill Lane crossing	CI	75	2,310	2029	Slip-lining
MR-0272	Hatherby	Market Place (south)	CI	250	2,310	2029	Slip-lining
MR-0273	Hatherby	High Causeway	CI	225	2,130	2029	Open cut
MR-0274	Hatherby	Hollow Way (east)	CI	90	1,270	2029	Open cut
MR-0275	Ivelstone	Ropewalk (east)	CI	225	1,790	2029	Slip-lining
MR-0276	Ivelstone	Beacon Rise (south)	GS	75	600	2029	Open cut
MR-0277	Ivelstone	Orchard Way	CI	75	1,830	2029	Pipe bursting
MR-0278	Ivelstone	Hollow Way	CI	150	390	2029	Slip-lining
MR-0279	Ivelstone	Station Road (west)	CI	75	780	2029	Open cut
MR-0280	Ivelstone	Hollow Way (south)	CI	125	2,310	2029	Slip-lining
MR-0281	Ivelstone	Tanners Row crossing	CI	75	1,930	2029	Directional drilling
MR-0282	Kelmarsh Vale	Market Place	PVC	450	70	2029	Open cut
MR-0283	Larkhollow	Beacon Rise (north)	CI	75	1,400	2029	Slip-lining
MR-0284	Larkhollow	Chapel Hill (south)	AC	180	630	2029	Pipe bursting
MR-0285	Larkhollow	Market Place (north)	PVC	75	830	2029	Pipe bursting
MR-0286	Larkhollow	Market Place crossing	CI	180	1,670	2029	Open cut
MR-0287	Larkhollow	Weavers Close (east)	AC	75	820	2029	Open cut
MR-0288	Larkhollow	Weavers Close to Long Furlong	GS	300	1,930	2029	Open cut
MR-0289	Larkhollow	Mill Lane (west)	AC	125	2,090	2029	Pipe bursting
MR-0290	Larkhollow	Hollow Way crossing	GS	150	1,390	2029	Pipe bursting
MR-0291	Larkhollow	Chapel Hill (east)	AC	300	120	2029	Pipe bursting
MR-0292	Marrowby	Quarry Lane (east)	GS	150	290	2029	Slip-lining
MR-0293	Marrowby	Quarry Lane	PVC	250	730	2029	Open cut
MR-0294	Marrowby	Long Furlong (west)	GS	180	470	2029	Open cut
MR-0295	Marrowby	Copse Road (west)	CI	125	1,300	2029	Pipe bursting
MR-0296	Marrowby	Bridge End crossing	CI	100	1,020	2029	Pipe bursting
MR-0297	Netherfold	Quarry Lane (south)	CI	450	2,190	2029	Open cut
MR-0298	Netherfold	Ferry Road (east)	DI	150	490	2029	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0299	Netherfold	Hollow Way (west)	CI	225	220	2029	Slip-lining
MR-0300	Netherfold	Mill Lane	GS	180	770	2029	Open cut
MR-0301	Netherfold	Quarry Lane to School Lane	AC	225	950	2029	Slip-lining
MR-0302	Netherfold	Orchard Way	AC	450	1,470	2029	Slip-lining
MR-0303	Netherfold	Ropewalk	DI	100	1,180	2029	Open cut
MR-0304	Netherfold	Kiln Street crossing	GS	150	100	2029	Pipe bursting
MR-0305	Netherfold	Kiln Street to Beacon Rise	GS	100	1,950	2029	Open cut
MR-0306	Netherfold	Mill Lane to Bridge End	CI	300	2,160	2029	Pipe bursting
MR-0307	Netherfold	Weavers Close (west)	CI	300	1,620	2029	Pipe bursting
MR-0308	Netherfold	High Causeway (west)	PVC	355	2,350	2029	Slip-lining
MR-0309	Pellow Bay	Hollow Way (west)	CI	300	1,480	2029	Slip-lining
MR-0310	Pellow Bay	Chapel Hill (west)	CI	225	610	2029	Pipe bursting
MR-0311	Pellow Bay	Orchard Way to Weavers Close	GS	180	2,330	2029	Open cut
MR-0312	Quarrendon	Station Road crossing	CI	75	2,190	2029	Directional drilling
MR-0313	Quarrendon	Gallows Hill (south)	DI	450	140	2029	Pipe bursting
MR-0314	Quarrendon	Gallows Hill (north)	CI	300	1,040	2029	Slip-lining
MR-0315	Quarrendon	Mill Lane (west)	CI	75	850	2029	Open cut
MR-0316	Quarrendon	Sandpit Lane crossing	CI	90	1,590	2029	Open cut
MR-0317	Quarrendon	Station Road (north)	GS	300	1,340	2029	Directional drilling
MR-0318	Quarrendon	Gallows Hill (east)	AC	180	290	2029	Slip-lining
MR-0319	Rushmere Down	Tanners Row (south)	PVC	355	690	2029	Directional drilling
MR-0320	Rushmere Down	High Causeway to Ropewalk	GS	300	1,480	2029	Open cut
MR-0321	Rushmere Down	Sandpit Lane crossing	DI	225	680	2029	Slip-lining
MR-0322	Rushmere Down	Sandpit Lane (east)	PVC	355	2,160	2029	Pipe bursting
MR-0323	Rushmere Down	Gallows Hill (south)	GS	300	2,220	2029	Directional drilling
MR-0324	Stavely Cross	Gallows Hill crossing	GS	450	260	2029	Open cut
MR-0325	Stavely Cross	Market Place	CI	300	510	2029	Slip-lining
MR-0326	Stavely Cross	Gallows Hill (north)	DI	225	210	2029	Open cut
MR-0327	Tolland Edge	Gallows Hill (north)	CI	180	650	2029	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0328	Tolland Edge	High Causeway (south)	PVC	100	1,020	2029	Open cut
MR-0329	Tolland Edge	Mill Lane (east)	CI	225	600	2029	Directional drilling
MR-0330	Tolland Edge	Gallows Hill (north)	CI	125	610	2029	Pipe bursting
MR-0331	Tolland Edge	Quarry Lane (north)	DI	100	510	2029	Open cut
MR-0332	Tolland Edge	Tanners Row crossing	DI	150	1,570	2029	Slip-lining
MR-0333	Ulverby	Mill Lane (east)	AC	355	1,530	2029	Open cut
MR-0334	Ulverby	Station Road (east)	CI	100	2,350	2029	Open cut
MR-0335	Ulverby	Copse Road	PVC	150	1,040	2029	Open cut
MR-0336	Vexford	Ferry Road	CI	75	1,300	2029	Directional drilling
MR-0337	Vexford	Bridge End crossing	GS	450	580	2029	Pipe bursting
MR-0338	Vexford	Bridge End	PVC	150	1,870	2029	Pipe bursting
MR-0339	Vexford	Mill Lane (south)	DI	125	2,340	2029	Open cut
MR-0340	Vexford	Bridge End (south)	CI	90	1,260	2029	Open cut
MR-0341	Wrayholme	School Lane crossing	DI	90	440	2029	Directional drilling
MR-0342	Wrayholme	Gallows Hill crossing	AC	100	1,140	2029	Slip-lining
MR-0343	Wrayholme	Chapel Hill (west)	CI	225	110	2029	Directional drilling
MR-0344	Wrayholme	Chapel Hill (west)	DI	90	870	2029	Open cut
MR-0345	Westcombe Rise	Ferry Road (east)	CI	450	110	2029	Slip-lining
MR-0346	Westcombe Rise	Market Place crossing	CI	250	1,740	2029	Pipe bursting
MR-0347	Westcombe Rise	School Lane	AC	75	2,340	2029	Pipe bursting
MR-0348	Westcombe Rise	Kiln Street (north)	PVC	150	190	2029	Slip-lining
MR-0349	Westcombe Rise	Gallows Hill (west)	AC	75	460	2029	Pipe bursting
MR-0350	Westcombe Rise	Weavers Close (south)	CI	355	1,890	2029	Open cut
MR-0351	Westcombe Rise	Hollow Way (east)	CI	125	1,550	2029	Directional drilling
MR-0352	Westcombe Rise	Tanners Row to Tanners Row	CI	250	850	2029	Pipe bursting
MR-0353	Westcombe Rise	Long Furlong (south)	CI	355	1,710	2029	Open cut
MR-0354	Yarrowfield	Bridge End (south)	AC	225	800	2029	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0355	Yarrowfield	Ropewalk to Bridge End	CI	180	80	2029	Directional drilling
MR-0356	Yarrowfield	Weavers Close (south)	AC	225	620	2029	Directional drilling
MR-0357	Yarrowfield	Orchard Way to Tanners Row	AC	180	1,780	2029	Directional drilling
MR-0358	Yeldham Moss	Ropewalk crossing	GS	125	480	2029	Directional drilling
MR-0359	Yeldham Moss	Mill Lane to Ferry Road	PVC	75	1,350	2029	Directional drilling
MR-0360	Yeldham Moss	Ropewalk	CI	90	1,800	2029	Slip-lining
MR-0361	Yeldham Moss	Station Road (south)	DI	225	360	2029	Slip-lining
MR-0362	Yeldham Moss	School Lane (south)	CI	180	470	2029	Open cut
MR-0363	Ardley Fen	Hollow Way crossing	DI	100	1,360	2030	Pipe bursting
MR-0364	Ardley Fen	Long Furlong	AC	225	320	2030	Directional drilling
MR-0365	Ardley Fen	Kiln Street to Beacon Rise	CI	100	400	2030	Directional drilling
MR-0366	Ardley Fen	Ropewalk	DI	90	1,330	2030	Slip-lining
MR-0367	Ardley Fen	Tanners Row crossing	AC	180	920	2030	Slip-lining
MR-0368	Brackwater	Hollow Way crossing	CI	225	2,100	2030	Open cut
MR-0369	Brackwater	Station Road to Gallows Hill	CI	125	1,620	2030	Pipe bursting
MR-0370	Brackwater	Station Road (north)	GS	300	520	2030	Open cut
MR-0371	Brackwater	Beacon Rise (south)	AC	450	2,150	2030	Slip-lining
MR-0372	Caddon Heath	Quarry Lane (west)	DI	180	1,710	2030	Open cut
MR-0373	Caddon Heath	Ferry Road	PVC	75	800	2030	Directional drilling
MR-0374	Caddon Heath	Ropewalk	PVC	300	740	2030	Open cut
MR-0375	Caddon Heath	Gallows Hill (south)	DI	150	250	2030	Open cut
MR-0376	Caddon Heath	Ropewalk crossing	AC	225	340	2030	Slip-lining
MR-0377	Caddon Heath	Beacon Rise (west)	CI	125	380	2030	Directional drilling
MR-0378	Caddon Heath	Ferry Road to High Causeway	CI	100	740	2030	Open cut
MR-0379	Caddon Heath	Orchard Way to Station Road	DI	125	1,950	2030	Directional drilling
MR-0380	Caddon Heath	Gallows Hill (north)	CI	150	90	2030	Pipe bursting
MR-0381	Dunholt	Market Place (west)	CI	250	1,520	2030	Directional drilling

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0382	Dunholt	Copse Road (east)	CI	90	720	2030	Pipe bursting
MR-0383	Dunholt	Hollow Way	CI	90	1,950	2030	Open cut
MR-0384	Dunholt	Copse Road (north)	DI	180	1,770	2030	Slip-lining
MR-0385	Dunholt	Chapel Hill (south)	CI	90	920	2030	Open cut
MR-0386	Dunholt	Ferry Road	DI	125	100	2030	Pipe bursting
MR-0387	Dunholt	Weavers Close crossing	CI	225	710	2030	Directional drilling
MR-0388	Dunholt	Long Furlong (north)	CI	300	490	2030	Directional drilling
MR-0389	Elmscar	High Causeway (east)	AC	450	1,290	2030	Slip-lining
MR-0390	Elmscar	Kiln Street (east)	CI	125	1,750	2030	Slip-lining
MR-0391	Elmscar	Quarry Lane (north)	CI	180	1,680	2030	Pipe bursting
MR-0392	Elmscar	Mill Lane (east)	GS	300	750	2030	Open cut
MR-0393	Fernlow	Sandpit Lane crossing	AC	150	310	2030	Slip-lining
MR-0394	Gorsemere	Hollow Way (south)	AC	225	440	2030	Open cut
MR-0395	Gorsemere	Hollow Way crossing	GS	355	2,060	2030	Pipe bursting
MR-0396	Gorsemere	Orchard Way (north)	CI	100	630	2030	Directional drilling
MR-0397	Hatherby	Ropewalk (west)	GS	100	390	2030	Open cut
MR-0398	Hatherby	High Causeway crossing	CI	150	1,620	2030	Pipe bursting
MR-0399	Hatherby	Market Place crossing	AC	150	1,810	2030	Slip-lining
MR-0400	Ivelstone	Bridge End to Beacon Rise	AC	450	840	2030	Open cut
MR-0401	Ivelstone	School Lane (east)	AC	180	1,700	2030	Open cut
MR-0402	Ivelstone	Kiln Street (south)	PVC	150	930	2030	Slip-lining
MR-0403	Kelmarsh Vale	Ropewalk crossing	AC	90	1,930	2030	Pipe bursting
MR-0404	Kelmarsh Vale	School Lane crossing	AC	450	1,380	2030	Open cut
MR-0405	Larkhollow	Station Road (south)	AC	300	1,670	2030	Pipe bursting
MR-0406	Larkhollow	Hollow Way to Tanners Row	CI	250	780	2030	Directional drilling
MR-0407	Marrowby	Mill Lane (south)	GS	450	1,620	2030	Pipe bursting
MR-0408	Marrowby	Ferry Road (south)	CI	150	1,490	2030	Open cut
MR-0409	Marrowby	Bridge End (east)	AC	125	1,850	2030	Open cut
MR-0410	Marrowby	Mill Lane (south)	CI	90	1,550	2030	Pipe bursting
MR-0411	Marrowby	Kiln Street	DI	180	940	2030	Slip-lining
MR-0412	Marrowby	Chapel Hill crossing	CI	75	1,680	2030	Directional drilling
MR-0413	Marrowby	Hollow Way (west)	CI	90	1,980	2030	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0414	Marrowby	Orchard Way crossing	DI	100	2,070	2030	Pipe bursting
MR-0415	Marrowby	Mill Lane (north)	GS	90	920	2030	Directional drilling
MR-0416	Marrowby	Copse Road	AC	225	520	2030	Directional drilling
MR-0417	Marrowby	Hollow Way (east)	PVC	300	2,030	2030	Pipe bursting
MR-0418	Netherfold	Quarry Lane	AC	100	750	2030	Pipe bursting
MR-0419	Netherfold	Hollow Way (west)	CI	450	140	2030	Open cut
MR-0420	Netherfold	Quarry Lane crossing	CI	180	1,520	2030	Open cut
MR-0421	Netherfold	Hollow Way (east)	CI	75	1,490	2030	Directional drilling
MR-0422	Netherfold	Gallows Hill to Copse Road	GS	355	900	2030	Directional drilling
MR-0423	Netherfold	Station Road (east)	CI	75	1,110	2030	Open cut
MR-0424	Netherfold	Ropewalk (east)	AC	300	1,210	2030	Slip-lining
MR-0425	Netherfold	Market Place	AC	250	2,340	2030	Open cut
MR-0426	Netherfold	Mill Lane crossing	AC	125	1,940	2030	Directional drilling
MR-0427	Netherfold	Hollow Way (south)	GS	450	1,490	2030	Open cut
MR-0428	Pellow Bay	Kiln Street crossing	GS	90	1,670	2030	Slip-lining
MR-0429	Pellow Bay	Sandpit Lane (south)	DI	150	310	2030	Directional drilling
MR-0430	Pellow Bay	Sandpit Lane (east)	CI	180	700	2030	Slip-lining
MR-0431	Pellow Bay	Orchard Way (south)	CI	450	2,320	2030	Open cut
MR-0432	Pellow Bay	Copse Road (east)	CI	180	1,620	2030	Pipe bursting
MR-0433	Quarrendon	Gallows Hill to Kiln Street	CI	125	960	2030	Slip-lining
MR-0434	Quarrendon	Hollow Way (west)	CI	100	530	2030	Slip-lining
MR-0435	Quarrendon	Chapel Hill to Market Place	CI	450	2,070	2030	Open cut
MR-0436	Quarrendon	Long Furlong to Market Place	CI	75	480	2030	Open cut
MR-0437	Quarrendon	Orchard Way (north)	PVC	355	90	2030	Open cut
MR-0438	Rushmere Down	Chapel Hill to Mill Lane	GS	355	2,040	2030	Pipe bursting
MR-0439	Rushmere Down	Quarry Lane (north)	AC	250	1,930	2030	Open cut
MR-0440	Rushmere Down	Mill Lane (east)	CI	180	2,130	2030	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0441	Rushmere Down	Market Place (south)	GS	250	1,270	2030	Open cut
MR-0442	Stavely Cross	Hollow Way (north)	DI	355	1,790	2030	Open cut
MR-0443	Stavely Cross	Weavers Close to Hollow Way	CI	250	120	2030	Slip-lining
MR-0444	Stavely Cross	Beacon Rise (east)	GS	250	670	2030	Slip-lining
MR-0445	Stavely Cross	Hollow Way (west)	CI	180	1,490	2030	Open cut
MR-0446	Tolland Edge	Beacon Rise (east)	PVC	90	500	2030	Slip-lining
MR-0447	Tolland Edge	Market Place to Bridge End	AC	150	1,060	2030	Open cut
MR-0448	Tolland Edge	Tanners Row crossing	CI	300	2,070	2030	Directional drilling
MR-0449	Tolland Edge	Hollow Way	CI	300	1,110	2030	Directional drilling
MR-0450	Tolland Edge	Orchard Way	CI	450	1,630	2030	Open cut
MR-0451	Tolland Edge	Kiln Street crossing	CI	450	390	2030	Open cut
MR-0452	Tolland Edge	Beacon Rise to School Lane	AC	75	1,370	2030	Directional drilling
MR-0453	Ulverby	Weavers Close (south)	DI	450	460	2030	Open cut
MR-0454	Ulverby	Orchard Way (south)	GS	150	440	2030	Pipe bursting
MR-0455	Ulverby	Copse Road	AC	180	1,230	2030	Open cut
MR-0456	Ulverby	Weavers Close (north)	DI	75	890	2030	Pipe bursting
MR-0457	Vexford	Ropewalk to Hollow Way	CI	300	150	2030	Directional drilling
MR-0458	Vexford	Market Place (north)	PVC	450	1,720	2030	Directional drilling
MR-0459	Vexford	Gallows Hill (south)	GS	90	520	2030	Open cut
MR-0460	Wrayholme	Hollow Way (north)	CI	180	1,670	2030	Open cut
MR-0461	Wrayholme	Hollow Way (north)	PVC	300	1,460	2030	Open cut
MR-0462	Westcombe Rise	Station Road	DI	100	340	2030	Pipe bursting
MR-0463	Westcombe Rise	Kiln Street (east)	CI	100	1,890	2030	Open cut
MR-0464	Westcombe Rise	Beacon Rise (south)	CI	225	1,840	2030	Directional drilling
MR-0465	Westcombe Rise	Long Furlong (east)	PVC	180	2,140	2030	Slip-lining
MR-0466	Westcombe Rise	School Lane	DI	180	530	2030	Open cut
MR-0467	Yarrowfield	Chapel Hill crossing	GS	450	1,370	2030	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0468	Yarrowfield	Long Furlong (east)	CI	250	1,220	2030	Open cut
MR-0469	Yarrowfield	Station Road crossing	AC	250	1,950	2030	Slip-lining
MR-0470	Yarrowfield	Orchard Way (east)	CI	90	920	2030	Pipe bursting
MR-0471	Yarrowfield	Quarry Lane (north)	CI	150	2,270	2030	Directional drilling
MR-0472	Yeldham Moss	Market Place (south)	CI	100	1,320	2030	Slip-lining
MR-0473	Yeldham Moss	Weavers Close to Orchard Way	DI	75	2,190	2030	Slip-lining
MR-0474	Yeldham Moss	Orchard Way to Gallows Hill	AC	90	2,240	2030	Pipe bursting
MR-0475	Yeldham Moss	Orchard Way (north)	CI	100	280	2030	Pipe bursting
MR-0476	Yeldham Moss	Beacon Rise (south)	CI	75	1,250	2030	Pipe bursting
MR-0477	Yeldham Moss	Bridge End to Ropewalk	AC	225	850	2030	Directional drilling
MR-0478	Ardley Fen	Sandpit Lane (east)	DI	355	2,060	2031	Slip-lining
MR-0479	Ardley Fen	Bridge End (west)	CI	250	1,780	2031	Slip-lining
MR-0480	Ardley Fen	Ferry Road (west)	CI	100	1,370	2031	Slip-lining
MR-0481	Ardley Fen	Beacon Rise (north)	DI	150	960	2031	Slip-lining
MR-0482	Ardley Fen	Hollow Way (north)	PVC	225	970	2031	Pipe bursting
MR-0483	Ardley Fen	High Causeway (south)	DI	225	860	2031	Slip-lining
MR-0484	Ardley Fen	Weavers Close (south)	CI	300	460	2031	Directional drilling
MR-0485	Brackwater	Long Furlong crossing	CI	450	910	2031	Open cut
MR-0486	Brackwater	School Lane (south)	CI	75	2,070	2031	Pipe bursting
MR-0487	Brackwater	Mill Lane (west)	PVC	450	1,990	2031	Slip-lining
MR-0488	Brackwater	Orchard Way crossing	AC	250	830	2031	Directional drilling
MR-0489	Brackwater	Bridge End (east)	PVC	180	550	2031	Open cut
MR-0490	Caddon Heath	Chapel Hill	AC	180	190	2031	Slip-lining
MR-0491	Caddon Heath	Tanners Row (east)	CI	100	700	2031	Pipe bursting
MR-0492	Caddon Heath	Sandpit Lane (south)	PVC	150	1,030	2031	Slip-lining
MR-0493	Caddon Heath	Ferry Road (north)	AC	100	1,900	2031	Pipe bursting
MR-0494	Dunholt	Gallows Hill (east)	PVC	450	2,330	2031	Pipe bursting
MR-0495	Dunholt	Kiln Street (south)	GS	75	650	2031	Directional drilling
MR-0496	Dunholt	Orchard Way (west)	AC	250	800	2031	Open cut
MR-0497	Elmscar	Long Furlong (east)	CI	100	880	2031	Slip-lining
MR-0498	Elmscar	Chapel Hill to Tanners Row	GS	225	1,670	2031	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0499	Elmscar	Weavers Close (east)	DI	90	1,050	2031	Directional drilling
MR-0500	Elmscar	Kiln Street crossing	AC	250	1,400	2031	Directional drilling
MR-0501	Fernlow	Weavers Close (north)	CI	355	2,030	2031	Directional drilling
MR-0502	Fernlow	Beacon Rise (east)	CI	180	1,680	2031	Slip-lining
MR-0503	Fernlow	Chapel Hill crossing	CI	225	1,980	2031	Slip-lining
MR-0504	Fernlow	High Causeway (south)	PVC	225	2,350	2031	Slip-lining
MR-0505	Fernlow	Tanners Row (south)	CI	225	1,520	2031	Open cut
MR-0506	Gorsemere	Ropewalk (north)	CI	250	830	2031	Open cut
MR-0507	Gorsemere	Quarry Lane to Station Road	CI	125	620	2031	Directional drilling
MR-0508	Gorsemere	Sandpit Lane crossing	CI	225	1,740	2031	Directional drilling
MR-0509	Gorsemere	Chapel Hill crossing	CI	450	150	2031	Directional drilling
MR-0510	Hatherby	Long Furlong (east)	PVC	300	930	2031	Open cut
MR-0511	Hatherby	Mill Lane (east)	PVC	225	1,490	2031	Open cut
MR-0512	Hatherby	Station Road (west)	AC	180	2,120	2031	Open cut
MR-0513	Hatherby	Hollow Way crossing	AC	150	1,660	2031	Pipe bursting
MR-0514	Hatherby	Weavers Close	CI	75	1,320	2031	Directional drilling
MR-0515	Hatherby	Station Road	PVC	75	240	2031	Pipe bursting
MR-0516	Hatherby	Market Place (south)	GS	450	270	2031	Slip-lining
MR-0517	Ivelstone	Kiln Street (west)	CI	450	380	2031	Open cut
MR-0518	Kelmarsh Vale	Market Place to Quarry Lane	DI	355	1,180	2031	Open cut
MR-0519	Kelmarsh Vale	Bridge End	AC	300	580	2031	Pipe bursting
MR-0520	Kelmarsh Vale	Copse Road	GS	75	2,060	2031	Directional drilling
MR-0521	Kelmarsh Vale	High Causeway to Tanners Row	DI	125	630	2031	Open cut
MR-0522	Larkhollow	Bridge End (east)	AC	450	1,860	2031	Open cut
MR-0523	Larkhollow	Gallows Hill crossing	CI	250	1,090	2031	Pipe bursting
MR-0524	Larkhollow	Bridge End crossing	GS	450	640	2031	Open cut
MR-0525	Larkhollow	Quarry Lane (south)	GS	180	670	2031	Pipe bursting
MR-0526	Larkhollow	Market Place crossing	CI	150	2,120	2031	Open cut
MR-0527	Larkhollow	Copse Road crossing	CI	300	450	2031	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0528	Larkhollow	Station Road to Tanners Row	GS	355	1,780	2031	Open cut
MR-0529	Larkhollow	Kiln Street to Beacon Rise	DI	125	2,280	2031	Open cut
MR-0530	Larkhollow	Ferry Road (south)	AC	125	720	2031	Open cut
MR-0531	Larkhollow	Copse Road to Beacon Rise	DI	225	2,210	2031	Open cut
MR-0532	Marrowby	Mill Lane to Long Furlong	CI	100	1,570	2031	Open cut
MR-0533	Marrowby	Station Road (west)	AC	125	1,640	2031	Directional drilling
MR-0534	Marrowby	Sandpit Lane crossing	AC	300	290	2031	Open cut
MR-0535	Marrowby	Weavers Close (north)	AC	75	1,300	2031	Pipe bursting
MR-0536	Marrowby	High Causeway to Beacon Rise	CI	90	2,340	2031	Open cut
MR-0537	Marrowby	Ropewalk (south)	AC	180	470	2031	Open cut
MR-0538	Marrowby	Bridge End (west)	GS	250	2,010	2031	Open cut
MR-0539	Marrowby	Tanners Row (south)	CI	75	2,340	2031	Open cut
MR-0540	Marrowby	Bridge End to Bridge End	CI	100	830	2031	Open cut
MR-0541	Marrowby	Kiln Street (east)	GS	450	230	2031	Slip-lining
MR-0542	Marrowby	Market Place (east)	DI	125	1,590	2031	Open cut
MR-0543	Netherfold	Bridge End crossing	CI	180	80	2031	Slip-lining
MR-0544	Netherfold	Hollow Way to Kiln Street	PVC	225	580	2031	Open cut
MR-0545	Netherfold	Quarry Lane crossing	AC	125	160	2031	Directional drilling
MR-0546	Pellow Bay	School Lane crossing	AC	150	1,240	2031	Open cut
MR-0547	Quarrendon	School Lane to Beacon Rise	CI	100	2,130	2031	Open cut
MR-0548	Quarrendon	Station Road to Market Place	PVC	100	310	2031	Open cut
MR-0549	Quarrendon	Ropewalk (east)	CI	250	1,940	2031	Directional drilling
MR-0550	Quarrendon	High Causeway (south)	CI	250	2,030	2031	Open cut
MR-0551	Quarrendon	Market Place (east)	CI	180	2,020	2031	Pipe bursting
MR-0552	Rushmere Down	Long Furlong (south)	AC	450	1,890	2031	Open cut
MR-0553	Rushmere Down	Mill Lane (south)	AC	450	720	2031	Pipe bursting
MR-0554	Rushmere Down	Gallows Hill (south)	DI	225	360	2031	Open cut

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0555	Rushmere Down	Ferry Road (east)	GS	225	940	2031	Open cut
MR-0556	Rushmere Down	Kiln Street (south)	AC	90	380	2031	Pipe bursting
MR-0557	Rushmere Down	Kiln Street (west)	AC	100	1,080	2031	Slip-lining
MR-0558	Rushmere Down	Long Furlong (south)	GS	450	1,030	2031	Directional drilling
MR-0559	Rushmere Down	Copse Road	CI	180	2,050	2031	Open cut
MR-0560	Stavely Cross	High Causeway (north)	AC	90	1,600	2031	Open cut
MR-0561	Stavely Cross	Market Place (north)	CI	180	1,590	2031	Open cut
MR-0562	Stavely Cross	Mill Lane (north)	DI	75	1,180	2031	Open cut
MR-0563	Stavely Cross	Market Place to Beacon Rise	AC	355	2,280	2031	Slip-lining
MR-0564	Stavely Cross	Beacon Rise (east)	CI	75	1,900	2031	Slip-lining
MR-0565	Stavely Cross	Weavers Close (east)	AC	125	1,450	2031	Directional drilling
MR-0566	Stavely Cross	Ropewalk (north)	CI	300	880	2031	Open cut
MR-0567	Stavely Cross	Copse Road	CI	450	1,060	2031	Open cut
MR-0568	Stavely Cross	Long Furlong (north)	DI	180	730	2031	Open cut
MR-0569	Tolland Edge	Hollow Way	CI	100	1,520	2031	Open cut
MR-0570	Tolland Edge	Ferry Road (south)	CI	300	2,110	2031	Open cut
MR-0571	Tolland Edge	Orchard Way (east)	CI	450	1,270	2031	Slip-lining
MR-0572	Tolland Edge	Beacon Rise (south)	DI	225	2,350	2031	Open cut
MR-0573	Ulverby	Long Furlong (north)	CI	150	1,410	2031	Directional drilling
MR-0574	Ulverby	High Causeway (east)	DI	450	620	2031	Open cut
MR-0575	Ulverby	High Causeway (south)	CI	100	1,510	2031	Open cut
MR-0576	Vexford	Mill Lane to Market Place	PVC	355	870	2031	Pipe bursting
MR-0577	Vexford	Sandpit Lane to Market Place	GS	225	1,890	2031	Open cut
MR-0578	Vexford	Ferry Road (east)	CI	100	1,120	2031	Open cut
MR-0579	Wrayholme	Kiln Street crossing	AC	225	690	2031	Open cut
MR-0580	Wrayholme	Sandpit Lane (west)	CI	150	2,300	2031	Directional drilling
MR-0581	Wrayholme	Quarry Lane (east)	CI	450	1,690	2031	Directional drilling
MR-0582	Wrayholme	Copse Road (north)	AC	225	420	2031	Slip-lining

Ref.	Zone	Location	Material	Diameter (mm)	Length (m)	Year	Method
MR-0583	Wrayholme	Chapel Hill (south)	GS	450	350	2031	Open cut
MR-0584	Wrayholme	Gallows Hill	GS	125	2,220	2031	Open cut
MR-0585	Wrayholme	Tanners Row (south)	CI	300	1,410	2031	Directional drilling
MR-0586	Westcombe Rise	Quarry Lane to Quarry Lane	PVC	75	2,280	2031	Pipe bursting
MR-0587	Westcombe Rise	High Causeway (west)	PVC	355	1,620	2031	Directional drilling
MR-0588	Westcombe Rise	Mill Lane (west)	CI	300	210	2031	Slip-lining
MR-0589	Yarrowfield	School Lane (north)	PVC	150	690	2031	Open cut
MR-0590	Yarrowfield	Hollow Way (north)	DI	150	250	2031	Pipe bursting
MR-0591	Yarrowfield	Mill Lane	AC	150	1,110	2031	Directional drilling
MR-0592	Yarrowfield	Kiln Street crossing	AC	250	1,060	2031	Slip-lining
MR-0593	Yarrowfield	Long Furlong to Ferry Road	CI	355	390	2031	Open cut
MR-0594	Yeldham Moss	Kiln Street (east)	DI	225	490	2031	Slip-lining
MR-0595	Yeldham Moss	Weavers Close (east)	PVC	450	1,290	2031	Open cut
MR-0596	Yeldham Moss	School Lane crossing	PVC	355	1,630	2031	Open cut
MR-0597	Yeldham Moss	Tanners Row (south)	CI	355	420	2031	Pipe bursting
MR-0598	Yeldham Moss	Kiln Street to High Causeway	AC	300	1,460	2031	Open cut
MR-0599	Yeldham Moss	Ferry Road	AC	180	1,250	2031	Directional drilling
MR-0600	Yeldham Moss	Beacon Rise (west)	CI	225	1,990	2031	Open cut

Appendix C Service reservoir schedule

All 61 service reservoirs, with their capacity, the year of construction, the condition grade from the 2025 survey and the action planned in this plan.

Reservoir	Zone	Capacity (Ml)	Built	Grade	Planned action
Ardley Fen No. 1	Ardley Fen	11.4	1883	1	Decommission after new storage is built
Ardley Fen No. 2	Ardley Fen	3.0	1921	4	Roof membrane renewal
Ardley Fen No. 3	Ardley Fen	18.2	1989	2	Roof membrane renewal
Brackwater No. 1	Brackwater	8.7	1940	1	Roof membrane renewal
Brackwater No. 2	Brackwater	9.6	1962	3	Roof membrane renewal
Brackwater No. 3	Brackwater	1.3	1895	3	Structural survey in 2028
Caddon Heath No. 1	Caddon Heath	12.5	1905	3	Replace with new tank on same site

Reservoir	Zone	Capacity (Ml)	Built	Grade	Planned action
Caddon Heath No. 2	Caddon Heath	7.6	1995	2	Internal coating and joint sealing
Caddon Heath No. 3	Caddon Heath	6.0	2010	4	Internal coating and joint sealing
Dunholt No. 1	Dunholt	11.6	1950	5	Internal coating and joint sealing
Dunholt No. 2	Dunholt	10.7	1957	3	Replace with new tank on same site
Dunholt No. 3	Dunholt	7.3	1993	2	No action in this plan
Elmscar No. 1	Elmscar	10.7	1887	3	Structural survey in 2028
Elmscar No. 2	Elmscar	1.2	1891	2	Roof membrane renewal
Elmscar No. 3	Elmscar	10.2	1894	4	Roof membrane renewal
Fernlow No. 1	Fernlow	10.3	1935	5	Roof membrane renewal
Fernlow No. 2	Fernlow	17.8	1973	4	No action in this plan
Fernlow No. 3	Fernlow	13.8	1888	2	Decommission after new storage is built
Gorsemere No. 1	Gorsemere	1.5	2001	5	No action in this plan
Gorsemere No. 2	Gorsemere	2.7	1915	3	Hatch and vent security upgrade
Gorsemere No. 3	Gorsemere	16.4	1998	2	Structural survey in 2028
Hatherby No. 1	Hatherby	8.6	2013	4	Hatch and vent security upgrade
Hatherby No. 2	Hatherby	5.4	1932	1	Hatch and vent security upgrade
Hatherby No. 3	Hatherby	2.7	1930	1	Decommission after new storage is built
Ivelstone No. 1	Ivelstone	15.9	1997	2	Internal coating and joint sealing
Ivelstone No. 2	Ivelstone	14.8	1955	1	Replace with new tank on same site
Ivelstone No. 3	Ivelstone	13.5	1889	2	Structural survey in 2028
Kelmarsh Vale No. 1	Kelmarsh Vale	6.3	1983	1	Internal coating and joint sealing
Kelmarsh Vale No. 2	Kelmarsh Vale	2.3	1995	2	Decommission after new storage is built
Kelmarsh Vale No. 3	Kelmarsh Vale	10.2	1980	1	Hatch and vent security upgrade
Larkhollow No. 1	Larkhollow	18.4	1972	5	Structural survey in 2028
Larkhollow No. 2	Larkhollow	11.7	1904	2	Replace with new tank on same site
Larkhollow No. 3	Larkhollow	7.5	1964	2	Replace with new tank on same site
Marrowby No. 1	Marrowby	1.7	1924	2	Roof membrane renewal
Marrowby No. 2	Marrowby	9.8	1975	3	No action in this plan
Marrowby No. 3	Marrowby	6.3	1979	5	Hatch and vent security upgrade
Netherfold No. 1	Netherfold	7.3	1938	5	Internal coating and joint sealing
Netherfold No. 2	Netherfold	6.9	1888	2	Internal coating and joint sealing
Netherfold No. 3	Netherfold	16.1	1916	2	Hatch and vent security upgrade
Pellow Bay No. 1	Pellow Bay	7.2	2007	5	Structural survey in 2028
Pellow Bay No. 2	Pellow Bay	6.3	1994	3	Roof membrane renewal
Quarrendon No. 1	Quarrendon	12.4	1940	1	Internal coating and joint sealing
Quarrendon No. 2	Quarrendon	2.0	1908	2	Hatch and vent security upgrade
Rushmere Down No. 1	Rushmere Down	9.4	1909	1	Internal coating and joint sealing

Reservoir	Zone	Capacity (Ml)	Built	Grade	Planned action
Rushmere Down No. 2	Rushmere Down	8.6	1937	1	Decommission after new storage is built
Stavely Cross No. 1	Stavely Cross	17.5	1996	1	Hatch and vent security upgrade
Stavely Cross No. 2	Stavely Cross	17.5	2009	2	No action in this plan
Tolland Edge No. 1	Tolland Edge	10.7	1981	1	Decommission after new storage is built
Tolland Edge No. 2	Tolland Edge	13.2	1974	5	Roof membrane renewal
Ulverby No. 1	Ulverby	9.7	1998	1	No action in this plan
Ulverby No. 2	Ulverby	14.6	1949	5	Roof membrane renewal
Vexford No. 1	Vexford	13.6	1975	5	Structural survey in 2028
Vexford No. 2	Vexford	18.3	2016	1	Hatch and vent security upgrade
Wrayholme No. 1	Wrayholme	5.0	1995	4	Roof membrane renewal
Wrayholme No. 2	Wrayholme	11.0	1929	4	Roof membrane renewal
Westcombe Rise No. 1	Westcombe Rise	16.8	1991	5	Structural survey in 2028
Westcombe Rise No. 2	Westcombe Rise	1.8	1906	5	No action in this plan
Yarrowfield No. 1	Yarrowfield	1.8	1891	1	Decommission after new storage is built
Yarrowfield No. 2	Yarrowfield	2.9	1949	1	No action in this plan
Yeldham Moss No. 1	Yeldham Moss	3.2	1909	5	Decommission after new storage is built
Yeldham Moss No. 2	Yeldham Moss	14.9	2006	3	Decommission after new storage is built

Appendix D Record of approval

This plan was prepared by the Asset Strategy Group and reviewed by the executive team before submission to the Board.

Step	Body	Date
Prepared	Asset Strategy Group	9 January 2027
Reviewed	Executive team	3 February 2027
Assured	Audit and risk committee	24 February 2027
Approved	Board of the Authority	18 March 2027

Synthetic sample. This document was generated for software testing and does not describe a real organisation.